

Statistical Learning for Music Genre Recognition

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Abstract

This work studies automatic music genre recognition from acoustic descriptors using both exploratory and supervised statistical learning methods. After preprocessing the data to reduce redundancy and improve numerical stability, principal component analysis is used to assess the intrinsic structure of the predictor space. The unsupervised results indicate that genre separation is only partial, suggesting that the available acoustic descriptors do not fully encode the complexity of musical genre boundaries. Supervised logistic models are then evaluated on a binary genre classification task, where they achieve strong predictive performance. The study also discusses the extension toward multinomial classification and the computational challenges associated with Newton-type estimation procedures.

I. INTRODUCTION

This project studies automatic musical genre recognition using a database composed of recordings from 60 performers. The material is organized into five categories, and for each performer, approximately 15 to 20 minutes of music were collected. These recordings were subsequently divided into 20 segments and parameterized.

The resulting descriptors combine standardized MPEG-7 audio features with cepstral coefficients and time-domain parameters. They summarize temporal, spectral, and envelope-related properties of each audio signal through statistical measures, such as mean values and variances computed either across frequency bands or over time.

The objective of this project is to study the information contained in the dataset through a descriptive analysis, followed by both univariate and multivariate analyses. Given the nature of the data, a filtering procedure is first implemented in order to select the most relevant and informative variables for the subsequent modeling stages.

After preprocessing, a principal component analysis (PCA) is proposed to explore the structure of the data in a reduced-dimensional space. In addition, an unsupervised classification approach based on hierarchical clustering with Ward's method is implemented, with the aim of comparing the obtained partitions with the true musical genre labels.

As a second component, supervised classifiers are constructed for the binary classification problem involving the *Classical* and *Jazz* genres. Several logistic regression models are considered: models including all available variables, models based on the most significant variables, and models obtained through stepwise variable selection. In order to compare their predictive performance, ROC curves are computed using the training data. The selection of a model of interest is then based on the area under the ROC curve (AUC), together with an analysis of the bias-variance trade-off associated with the implemented models.

Regularization strategies are also introduced through ridge logistic regression. In this case, the regularization parameter is estimated by means of 10-fold cross-validation on the training sample.

In the final part, the binary classification framework is extended to the nominal multinomial case in order to handle the five musical genres simultaneously. The multinomial logistic model is first formalized by choosing one genre as the reference category and expressing the class probabilities through a softmax-type link function. The likelihood, gradient, and Hessian are then derived to describe how the model parameters can be estimated numerically. A Newton-based implementation is critically analyzed, with particular attention to the numerical stability of the Hessian computation. Finally, the model is estimated using standard R functions, its performance is evaluated on both training and test data, and the multiclass results are interpreted through one-versus-rest ROC curves.

II. DATA DESCRIPTION AND PREPROCESSING

A. Data description

As previously described in detail, the main dataset contains 7,773 labeled tracks and 192 columns in total: 191 acoustic descriptors and the response variable **GENRE**. The five musical genres represented in the study are *Blues* with 1,074 tracks, *Classical* with 2,318 tracks, *Jazz* with 2,036 tracks, *Pop* with 1,033 tracks, and *Rock* with 1,312 tracks. This class distribution is summarized in Fig. 1.

The companion holdout file contains 3,798 additional tracks with the same 191 predictors, but without genre labels.

The dataset contains no missing values. The 191 numerical predictors are grouped into several families of acoustic descriptors, including MPEG-7-based spectral features, audio spectrum envelope summaries, spectral flatness measures, MFCC coefficients, and dedicated time-domain descriptors related to RMS energy, threshold crossings, and zero-crossing behavior.

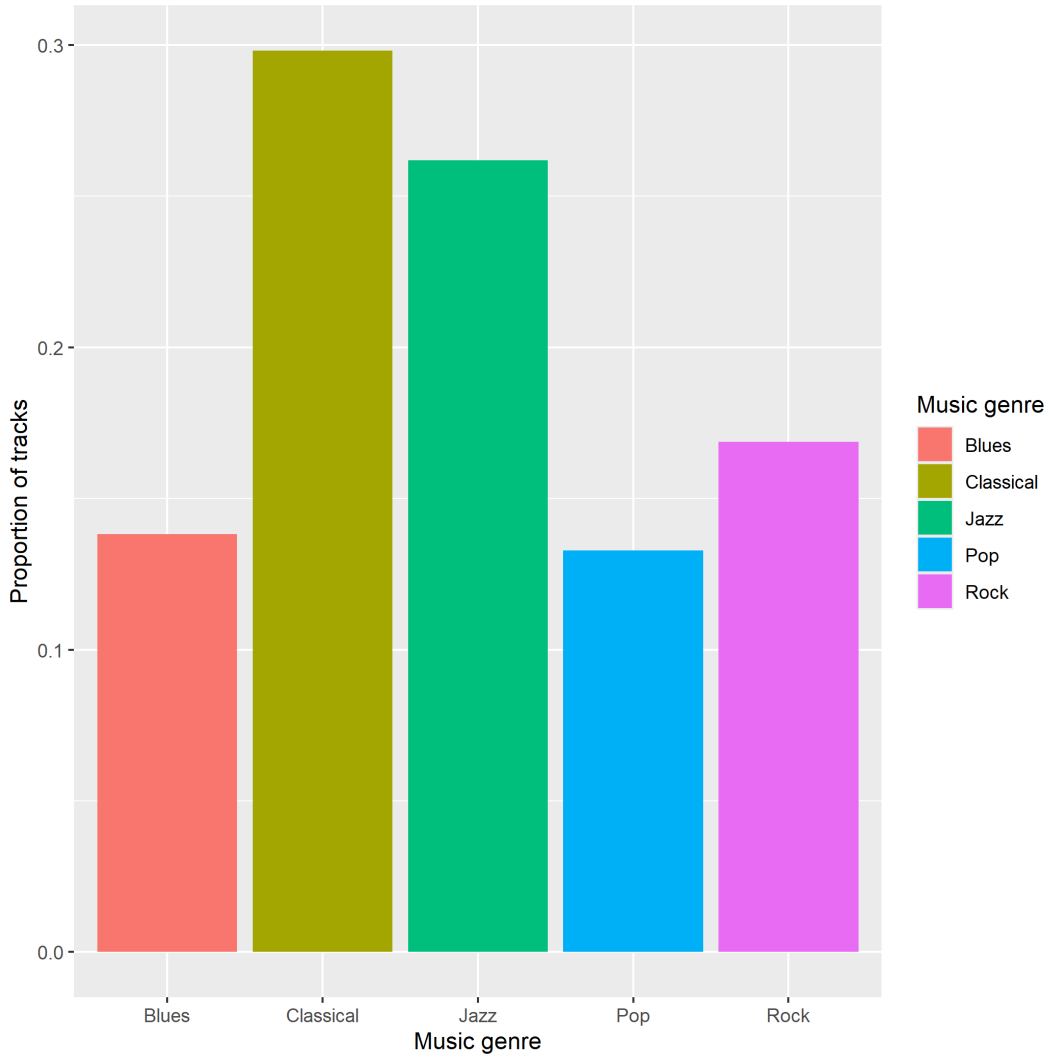


Fig. 1. Observed proportions of the five genres in the labeled dataset. Classical and Jazz are the two most represented classes.

B. Data Preprocessing

[1] discusses the possibility of applying a logarithmic transformation when large differences in scale are observed among variables characterized by strong asymmetry. This transformation is useful because it reduces large values more strongly than small values, which helps stabilize dispersion and prevents the analysis from being dominated by variables with higher magnitudes.

In order to identify the variables in the database that satisfy these dispersion-related conditions and may therefore benefit from the application of a logarithmic transformation, several complementary analyses are proposed. First, the mean-to-median ratio is computed as a simple descriptive indicator of asymmetry. For a roughly symmetric variable, the mean and the median are expected to be close, so the ratio should be near 1. A ratio clearly larger than 1 suggests right-skewness, since a small number of large observations pulls the mean upward relative to the median. The corresponding results are shown in the supplementary figure 4.

This analysis is further complemented by the introduction of a skewness estimate, defined as the third central moment of each variable. This measure provides a more detailed characterization of the asymmetry of the corresponding distribution. Large positive skewness values indicate the presence of a long right tail, whereas values close to zero are associated with a more symmetric distribution. The mathematical definition of skewness is presented in Appendix B, specifically in Equation (9). In addition, Fig. 5 shows the main candidate variables for the application of a logarithmic transformation according to their skewness values.

As a final complementary diagnostic, we examine heteroscedasticity in the conceptual sense of a *spread-versus-*

level pattern, that is, a situation in which the dispersion of a variable tends to increase with its magnitude instead of remaining approximately stable. This criterion is relevant here because a logarithmic transformation is especially useful when large values are not only more frequent in the tail, but also more variable. For that reason, an empirical heteroscedasticity score is used together with the mean-to-median ratio and the skewness coefficient to identify plausible candidates for logarithmic transformation. The complete definition of this score and its theoretical motivation are given in Appendix B, Subsection B-C; the corresponding ranking of variables is displayed in Fig. 6.

Nevertheless, before applying the transformation, it is important to consider the observation made by [2] regarding the logarithmic transformation: it does not necessarily guarantee symmetry. The improvement in terms of asymmetry reduction, approximation to normality, and variance stabilization is favored when the original data distribution is close to a log-normal distribution. For this reason, in order to assess whether a candidate variable can be reasonably considered approximately log-normal, a Shapiro–Wilk normality test is applied to its logarithmic transformation, considering only variables with strictly positive values. This criterion is based on the fact that, if a variable X follows a log-normal distribution, then $\log(X)$ should approximately follow a normal distribution. The common candidate set is listed in Appendix B, Subsection B-D, Table V, while the corresponding Shapiro–Wilk rankings are reported in Fig. 7 and Table VI.

Considering the 30 variables showing the strongest suitability as candidates for the application of a logarithmic transformation, the variables `PAR_SC_V` and `PAR_ASC_V` are common to the lists of the 30 best candidates according to the previously described criteria; see Appendix B, Table V. After applying the Shapiro–Wilk normality test to the logarithmically transformed variables, `PAR_SC_V` is found to be the variable that best follows the desired distributional behavior, in accordance with the recommendations of [2]; this ranking is summarized in Fig. 7 and Table VI. This makes `PAR_SC_V` the strongest candidate for the application of the logarithmic transformation. Additionally, `PAR_ASC_V` appears consistently among the variables of interest and is also included among the variables for which the Shapiro–Wilk test suggests a distribution more compatible with obtaining favorable results after the logarithmic transformation. Therefore, as part of the preprocessing stage and before centering and scaling the data, the logarithmic transformation is applied to `PAR_SC_V`, which is the strongest candidate. In addition, `PAR_ASC_V` is also included as a weaker but still relevant candidate, since its empirical distribution remains suitable for the application of this transformation. Consequently, both transformations are applied to the training data.

From the multivariate analysis, the inspection of the correlation matrix among the variables in the dataset reveals a second important preprocessing issue: redundancy.

A total of 22 variable pairs with absolute correlation greater than 0.99 are detected. Among them, 20 correspond to exact duplicates of the form $(\text{PAR_MFCC}j, \text{PAR_MFCCV}j)$, for $j = 1, \dots, 20$. This result is consistent with the duplicated block reported in variables 148 to 167 of the project statement.

The two remaining highly correlated pairs are $(\text{PAR_ZCD}, \text{PAR_ZCD_10FR_MEAN})$, with a correlation of 0.9958, and $(\text{PAR_1RMS_TCD}, \text{PAR_1RMS_TCD_10FR_MEAN})$, with a correlation of 0.9938. To reduce redundancy and avoid introducing duplicated information into the subsequent analyses, one representative variable is retained from each pair, while the corresponding redundant companion variable is removed.

The group of descriptor variables `PAR_ASE_M`, `PAR_ASE_MV`, `PAR_SFM_M`, and `PAR_SFM_MV` is also removed. These variables are computed as averages over families of variables that are already included in the dataset. Therefore, retaining both the detailed band-level variables and their aggregated summaries would introduce strong collinearity into the predictor space. This redundancy could make model estimation less stable, complicate the interpretation of the results, and lead to artificial overrepresentation of descriptor families that are already accounted for by their individual components. For this reason, these aggregated variables are excluded from the subsequent analyses. After all filtering steps, the predictor space is reduced from 191 to 165 variables.

III. UNSUPERVISED ANALYSIS

A. Principal Component Analysis

The preprocessed data are then centered and scaled. Principal component analysis (PCA) is performed on the centered and scaled matrix of the 165 retained predictors, so that the decomposition is driven by the correlation structure rather than by the original measurement scales.

If λ_k denotes the k -th eigenvalue of the correlation matrix, the proportion of explained inertia associated with axis k is given by

$$\frac{\lambda_k}{\sum_{j=1}^{165} \lambda_j}. \quad (1)$$

Numerically, the first two principal components explain 29.79% of the total inertia, the first four explain 43.52%, the first ten explain 59.01%, and the first 34 explain 81.60%; these values are summarized in Appendix B, Subsection B-E,

Table VII. Therefore, PCA reveals a nontrivial low-dimensional structure in the data. However, a substantial part of the total variability remains distributed across many directions, which limits the possibility of obtaining a clearly separated five-class visualization in only two dimensions. This limitation can also be observed by analyzing the quality of representation of the individuals. For an observation i , the quality of representation on the first k principal axes is measured by the cumulative squared cosine:

$$\cos_i^2(1:k) = \sum_{s=1}^k \cos_i^2(s). \quad (2)$$

The mean value of this quantity is 0.2888 on the first two axes, 0.4283 on the first four axes, 0.5900 on the first ten axes, and 0.8064 on the first 34 axes; see again Table VII. Hence, the first principal plane captures less than one third of the total information for a typical observation, whereas a substantially better representation is obtained only when considering a higher-dimensional subspace. A similar limitation is observed in the variable space. The analysis of the cumulative squared cosines shows that only a limited subset of variables is well represented on the first principal plane; see Appendix B, Tables XII and XIV. Although the representation improves when the first four principal axes are considered, a large part of the predictor set still requires a higher-dimensional subspace to be properly projected; see Tables XIII and XIV. Consequently, the interpretation of the PCA should not rely exclusively on the first two-dimensional representation, but should also account for higher-dimensional principal subspaces. In the first principal plane, the variables with the largest contributions include several descriptors from the spectrum envelope family; see Appendix B, Table VIII. This suggests that the first principal axis represents a strong contrast mainly associated with these descriptors.

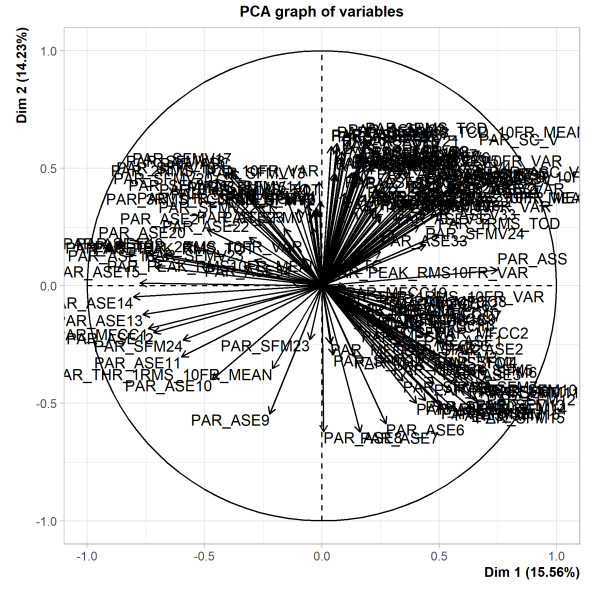
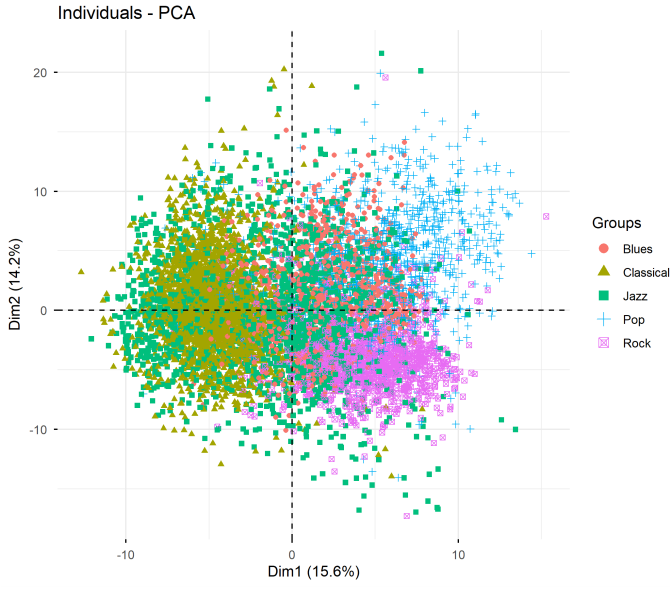
On the second principal axis, the most contributive variables are dominated by variability descriptors and by variables related to temporal features, such as threshold and zero-crossing descriptors; see Table IX. Therefore, the second axis of the principal plane can be interpreted as being more clearly associated with temporal crossing features and variability-related descriptors.

The relationships between the principal axes of the second plane and the variables of interest are weaker. Nevertheless, the third principal axis still shows an association with temporal crossing features and variability descriptors; see Table X. In contrast, the fourth principal axis appears to represent an opposition between certain ASE variables and SFM/SFMV variables, that is, between part of the audio spectrum envelope information and part of the spectral flatness information; see Table XI.

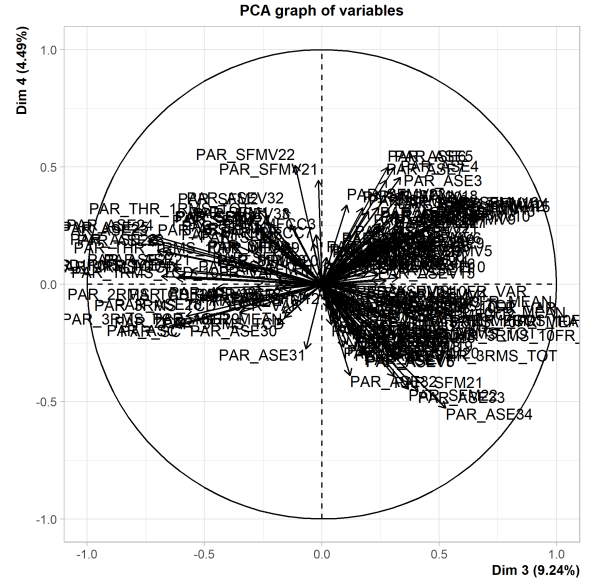
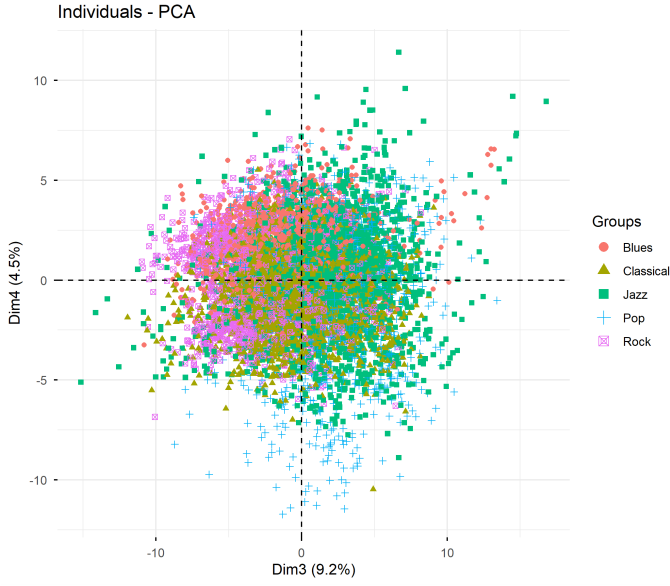
The first two principal planes are displayed jointly in Fig. 2. From these figures and from the results discussed above, it can be concluded that the first two principal planes are not sufficient to provide an adequate representation of either the individuals or the variables. Although the dataset exhibits a nontrivial underlying structure, this structure is not strongly concentrated in a very low-dimensional subspace. The first principal plane, shown in panel (a) of Fig. 2, reveals a partial horizontal contrast. *Pop* observations are more frequently located on the positive side of the first principal component, whereas *Classical* observations tend to be more concentrated on the negative side. *Rock* also tends to occupy the lower-right region of the plane. However, the point clouds remain broad and strongly overlapping, indicating that the first principal plane does not provide a sharp discrimination among the five musical genres. The second principal plane, shown in panel (b) of Fig. 2, provides additional information, but it does not eliminate the overlap between genres. Overall, the PCA confirms that the dataset contains relevant multivariate structure. Nevertheless, this structure is not organized into five clearly separated linear groups, which limits the discriminative power of a purely two-dimensional PCA visualization.

B. Hierarchical Clustering and Silhouettes

For the implementation of the unsupervised classifier, Ward’s hierarchical clustering is applied to the PCA scores. Several reduced representations are considered by retaining $d \in \{4, 6, 10, 34, 165\}$ principal components. The choice $d = 4$ corresponds to the first two principal planes. The values $d = 6$ and $d = 10$ are included because the scree plot of the percentage of inertia explained by each component shows relevant breaks before regions where the marginal gains in inertia become more homogeneous. Finally, $d = 34$ is considered because it corresponds to the set of principal components whose individual contribution remains above the average contribution expected under a uniform distribution of inertia across the variables and $d = 165$ corresponds to all of the components. For each retained dimension d , the number of clusters k is then chosen by maximizing the average silhouette width over $k \in \{2, \dots, 10\}$; the silhouette criterion is recalled in Appendix B, Subsection B-F, Eq. (11). In addition, the effect of unsupervised classification methods is also evaluated in the PCA-derived spaces after normalizing the retained principal component scores before applying hierarchical clustering. This complementary strategy makes it possible to assess whether differences in the scale of the selected PCA coordinates affect the resulting partitions.



(a) First principal plane (PC1, PC2).



(b) Second principal plane (PC3, PC4).

Fig. 2. Composite representation of the two principal planes used in the unsupervised analysis.

TABLE I
AVERAGE SILHOUETTE WIDTHS FOR WARD CLUSTERING AND FOR THE TRUE GENRE PARTITION.

Mode	PCs	Selected k	Ward silhouette	Genre silhouette
Not normalized	4	3	0.2170	0.0725
Not normalized	6	3	0.2029	0.0739
Not normalized	10	2	0.1644	0.0769
Not normalized	34	2	0.1212	0.0563
Not normalized	165	2	0.1059	0.0403
Normalized	4	3	0.1943	0.0604
Normalized	6	3	0.1283	0.0574
Normalized	10	5	0.0902	0.0532
Normalized	34	3	0.8171	-0.0062
Normalized	165	2	0.6843	-0.0435

The corresponding results are summarized in Table I. For most dimensions, the Ward partition achieves only moderate

average silhouette widths, while the silhouette values induced by the true genre labels remain low, which confirms that the five genres are only weakly separated in these reduced Euclidean representations.

The results make it clear that the partition of the true musical genres in the spaces constructed from the retained PCA variables is weak. The genre-wise silhouette values are low for both normalized and non-normalized representations, even reaching negative average values in the normalized case with $d = 34$. This indicates that the musical genres do not form compact or easily separable groups in the Euclidean spaces and geometries considered. Moreover, this behavior is not homogeneous across genres: although overlap is present for all genres, it is more pronounced for some of them. Therefore, the results obtained with hierarchical clustering not only fail to reproduce the true genre labels, but also reveal that the unsupervised structure preferred by the dataset is different from the five-genre partition. Using the implemented criterion for selecting the number of clusters, the preferred solution is almost always $k = 2$ or $k = 3$. The only case in which $k = 5$ is selected occurs for the normalized representation with $d = 10$, but the corresponding mean silhouette remains very low, with a value of 0.0902. Even for the preferred unsupervised partitions, the Ward silhouettes remain weak, which indicates the presence of some structure, but with relatively poor separation between groups.

It is also important to note that the normalized cases with $d = 34$ and $d = 165$ lead to degenerate partitions. For $d = 34$, the clustering produces one very large cluster containing 7,766 observations and two extremely small clusters of sizes 3 and 4, while for $d = 165$ it produces one cluster of size 7,770 and one cluster of size 3. These solutions are therefore not meaningful for interpretation, despite their large average Ward silhouettes. In general, normalization tends to reduce the average silhouette obtained by the unsupervised method when compared with the non-normalized PCA representations. This effect may be associated with the fact that normalization assigns equivalent weight to axes that do not contain the same amount of data variability. Since the first principal components are precisely those that concentrate the dominant structure, reweighting all retained axes equally can alter the distances between observations and increase the overlap between the hierarchical structures identified by the clustering algorithm. Overall, it can be concluded that the dataset does contain multivariate structure. However, this structure appears to be organized into a small number of broad and not fully separated groups, rather than into five well-defined clusters aligned with the musical genres.

C. Train Test split

The project includes an unlabeled holdout file. However, the project statement also requires an internal train–test split within the labeled sample. Therefore, the prescribed random split is applied, assigning observations to the training set with probability 2/3 and to the test set with probability 1/3, using `set.seed(103)` for reproducibility.

The binary *Classical* versus *Jazz* subsets derived from this split are used in Section IV.

IV. BINARY CLASSIFICATION

This section focuses on the binary discrimination between *Classical* and *Jazz*, which are the two largest classes in the labeled sample. Restricting the dataset to these two genres yields 4,354 observations in total. After applying the prescribed split described in Section II, the training subset contains 2,851 tracks, including 1,512 *Classical* tracks and 1,339 *Jazz* tracks. In the binary response variable, *Jazz* is encoded as the success category.

A. Logistic regression models

Given the nature of the problem, a logistic regression model is proposed in order to classify an observation x , defined in the space of the selected system variables, into the category of interest, which has been fixed as *Jazz*. Therefore, the mathematical formulation used for the model is

$$P(Y = 1 \mid X = x) = \frac{\exp(\beta_0 + x^\top \beta)}{1 + \exp(\beta_0 + x^\top \beta)}. \quad (3)$$

However, the predictor space has a relatively high dimension, with 165 retained predictors. For this reason, several maximum-likelihood logistic regression models are compared, each imposing a different restriction on the number of retained variables. The models are defined as follows:

- **Full logistic model (ModT):** includes all 165 retained predictors.
- **Mod1:** retains only predictors whose Wald p-value satisfies $p_j < 0.05$ in ModT.
- **Mod2:** retains only predictors whose Wald p-value satisfies $p_j < 0.20$ in ModT.
- **Stepwise logistic model (ModAIC):** selects variables by applying `stepAIC` to ModT.

This comparison makes it possible to evaluate the effect of variable selection on model complexity, numerical stability, interpretability, and predictive performance.

The relative adequacy of the reduced unpenalized logistic models with respect to the full model is assessed through likelihood-ratio tests based on deviance differences. The corresponding results are reported in Table II.

TABLE II
ADEQUACY TESTS OF REDUCED LOGISTIC MODELS AGAINST THE FULL MODEL

Model	Df	Deviance	p -value	Adequate
Mod1	107	256.51	2.70×10^{-14}	No
Mod2	83	109.65	0.0267	No
ModAIC	74	37.01	0.9999	Yes

The reduced models are evaluated through adequacy tests in order to determine whether a simpler specification can preserve the essential fit of the full logistic model. In logistic regression, this is assessed by comparing nested models using the difference in deviances, which measures the gain in log-likelihood obtained when moving from the reduced model to the complete model.

As shown in Fig. II, only the stepwise AIC model exhibits an adequate behavior. The deviance-based likelihood-ratio comparison indicates that **ModAIC** is not significantly inferior to **ModT**, suggesting that the AIC-selected model retains the essential fit of the full specification while using fewer predictors.

B. Regularized Logistic Regression

The interest of ridge regression in this study comes from the combination of a high-dimensional predictor space and residual dependence between acoustic descriptors, even after the preprocessing stage. In such a setting, an unpenalized logistic model may yield unstable coefficient estimates and may overfit the training data. Ridge regularization addresses this issue by shrinking the coefficients toward zero, which improves numerical stability and reduces the variance of the fitted classifier.

More specifically, ridge is useful here because it allows the model to keep the information carried by many correlated predictors without relying on a hard variable-selection rule. Instead of arbitrarily retaining only one variable among several related descriptors, the penalty distributes the effect more smoothly across them. For this reason, ridge is introduced as a way to control effective model complexity and to improve generalization performance in the binary classification task. The corresponding mathematical formulation is recalled in Appendix B, Subsection B-J.

In the **glmnet** algorithm, two extreme values of the regularization parameter are considered. The first one corresponds to $\lambda = 10^{10}$, a case in which the penalization term dominates the fitting criterion. As a consequence, the coefficients associated with the predictors are strongly shrunk and approach values close to zero at the end of the training process. The second case corresponds to $\lambda = 10^{-2}$, where the penalization is weak. In this situation, the fitted model is closer to a non-regularized logistic regression model using all retained predictors. Therefore, this configuration can be interpreted as being conceptually close to the maximum-likelihood logistic model.

Ridge regularization does not produce a single model directly, but rather a complete regularization path, corresponding to the family of fitted models obtained for different values of λ . However, from this family of models, it is necessary to select one final specification associated with a particular value of the regularization parameter.

The method proposed for selecting this value of λ is 10-fold cross-validation. The training set is divided into 10 subsamples. Successively, 9 of them are used to fit the ridge model, while the remaining subsample is used to evaluate the prediction error. This procedure is repeated until each fold has been used once as the validation sample.

For each value of λ in the grid, the prediction errors obtained over the 10 folds are averaged. The final retained value is **lambda.min**, that is, the value of λ that minimizes the mean cross-validation error. This leads to the final regularized model used for the binary classification task.

From the cross-validation results, displayed in Fig. 3, it can be observed that when λ is very large, the penalization is too strong. As a consequence, the model becomes highly constrained and the binomial deviance remains high. Conversely, as λ decreases, the model gains flexibility and the error decreases, indicating that, for the training data, an excessively strong penalization deteriorates the fit.

This suggests that, in order to minimize the binomial deviance, the model needs to remain relatively close to the non-regularized logistic regression. However, since ridge regression retains the 165 predictors and does not perform hard variable selection, the signal is still distributed across many predictors, including weak or redundant ones.

C. Summary of Binary Classification Results

Table III summarizes the main performance indicators for all five models evaluated on the test set. The AUC values are uniformly high, reflecting the strong linear separability of *Classical* and *Jazz* in the acoustic feature space. All models outperform the random baseline by a large margin.

The ridge model achieves a competitive AUC despite using a soft rather than hard variable-selection strategy, confirming that regularization provides a meaningful alternative to coefficient-wise significance thresholding when many predictors are correlated.

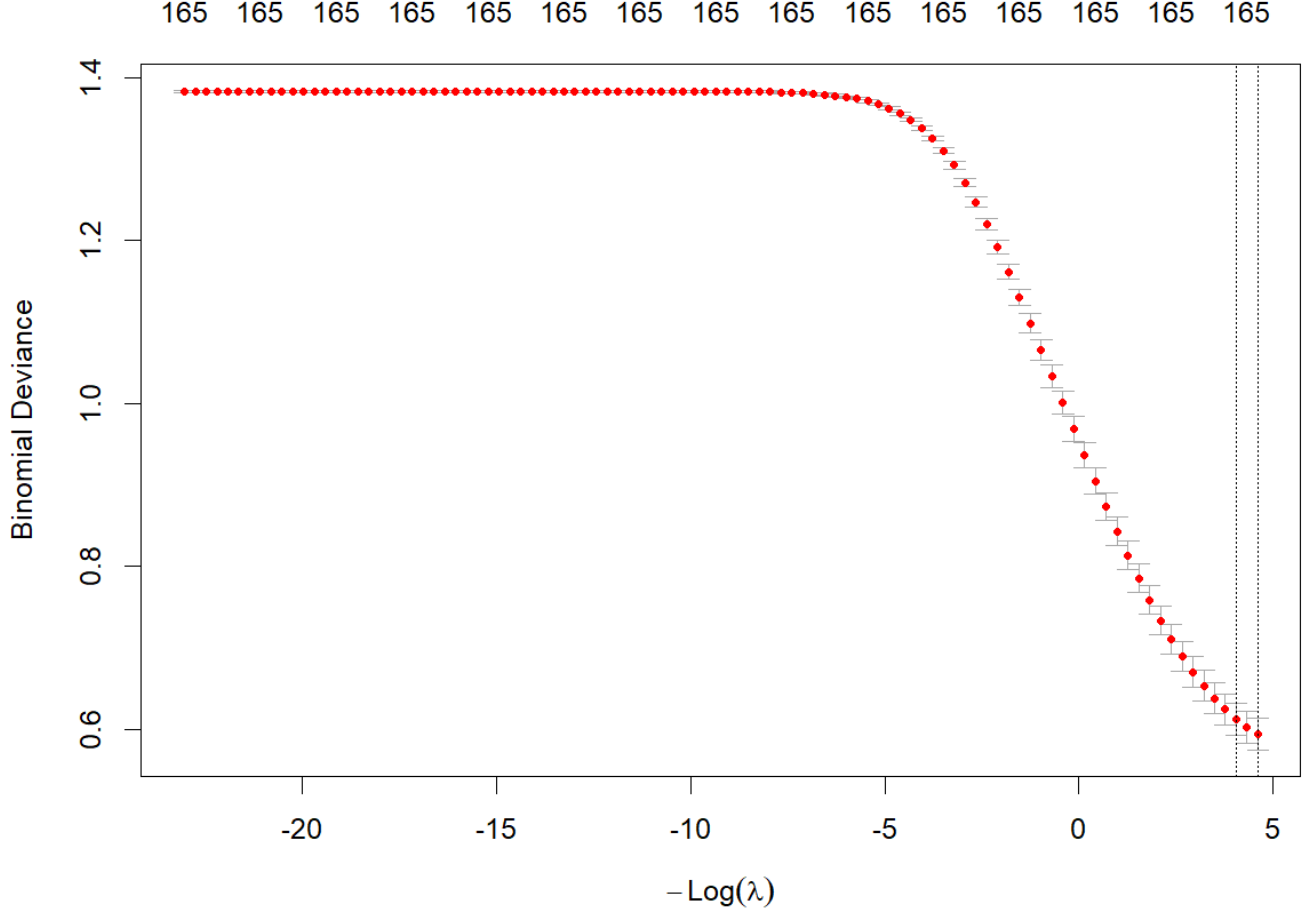


Fig. 3. Ten-fold cross-validation curve used to select the ridge regularization parameter. The retained value corresponds to the minimizer of the mean binomial deviance.

TABLE III
PERFORMANCE SUMMARY FOR ALL BINARY CLASSIFIERS ON THE TEST SET.

Model	AUC (test)	Accuracy	Error rate	# Variables
ModT	0.9627	0.9095	0.0905	165
Mod1	0.9543	0.8902	0.1098	58
Mod2	0.9605	0.9022	0.0978	82
ModAIC	0.9629	0.9082	0.0918	91
Ridge	0.9414	0.8842	0.1158	165

Based on the empirical evaluation on the test set, **ModAIC** achieves the highest AUC (0.9629) and the second highest accuracy (0.9082). It also provides a more parsimonious representation than the full and ridge models by reducing the active predictor space from 165 to 91 variables. Consequently, **ModAIC** is selected as the final binary classifier for this task. The genre predictions for the unlabeled holdout file were generated using this model and saved to the corresponding submission file.

V. MULTINOMIAL CLASSIFICATION

The binary framework is now extended to the full five-genre problem. Each observation is modeled as a realization of a multinomial distribution with $C = 5$ categories, and the softmax link function is used to relate the class probabilities to the linear predictors.

TABLE IV
TRAINING AND TEST ERROR RATES FOR THE MULTINOMIAL LOGISTIC MODEL.

Dataset	Error rate
Training	0.0681
Test	0.1022

A. Exponential Family and Softmax Link

The multinomial distribution belongs to the exponential family with natural parameter $\eta = (\log(\pi_1/\pi_C), \dots, \log(\pi_{C-1}/\pi_C))$, where π_c denotes the probability of class c and class C serves as the reference. Conditioning on a covariate vector x and defining the linear predictor $\eta^{(c)}(x) = x\theta^{(c)}$ for $c = 1, \dots, C-1$, the class probabilities are recovered through the softmax function:

$$\pi_c(x) = \frac{\exp(x\theta^{(c)})}{1 + \sum_{j=1}^{C-1} \exp(x\theta^{(j)})}, \quad c = 1, \dots, C-1, \quad (4)$$

$$\pi_C(x) = \frac{1}{1 + \sum_{j=1}^{C-1} \exp(x\theta^{(j)})}. \quad (5)$$

B. Log-Likelihood and Newton Estimation

For a sample of K independent observations, the log-likelihood is

$$\mathcal{L}(\theta) = \sum_{k=1}^K \sum_{c=1}^C y_{kc} \log \pi_c(x_k), \quad (6)$$

where $y_{kc} = \mathbf{1}[Y_k = c]$. The gradient with respect to $\theta_j^{(c)}$ is

$$\frac{\partial \mathcal{L}}{\partial \theta_j^{(c)}} = \sum_{k=1}^K x_{kj} (y_{kc} - \pi_{kc}), \quad (7)$$

and the Hessian block for components (j, c) and (l, d) is

$$\frac{\partial^2 \mathcal{L}}{\partial \theta_j^{(c)} \partial \theta_l^{(d)}} = - \sum_{k=1}^K x_{kj} x_{kl} \pi_{kc} (\mathbf{1}_{c=d} - \pi_{kd}). \quad (8)$$

Since the model belongs to the exponential family, the observed Hessian in Eq. (8) coincides with the expected Fisher information matrix. Consequently, Newton's method and the IRLS algorithm (Fisher scoring) produce exactly the same sequence of iterates in this setting. The suggestion made by the generative tool examined in Appendix A to replace Newton by IRLS for numerical stability is therefore misleading: the two methods are mathematically equivalent here, and the instability observed in practice is attributable to implementation details rather than to a conceptual difference between the algorithms.

C. Estimation with *multinom*

The multinomial logistic model is estimated using `nnet::multinom`, which internally maximizes the log-likelihood in Eq. (6) through a quasi-Newton optimizer with a cross-entropy objective. The training and test sets used here include all five genres, derived from the same split defined in Section II. The model is fitted on the training set and evaluated on the test set through the overall misclassification rate and the confusion matrix. The training and test error rates are reported in Table IV.

D. Neural Network Interpretation

The function `multinom` is implemented through a call to `nnet` with no hidden layer (`size = 0`), direct input-to-output connections (`skip = TRUE`), and softmax output activation. This architecture corresponds to a single-layer feedforward network with p input units and C output units connected by a softmax function, which is equivalent to the multinomial logistic model. Calling `nnet` directly with the same initialization (`rang` proportional to the data scale) and no weight decay (`decay = 0`) reproduces the predictions of `multinom` up to numerical precision, confirming the equivalence between the two parameterizations.

E. One-versus-Rest ROC Curves

In the multiclass setting, a single ROC curve is no longer well-defined, because the ROC framework requires a binary partition into positive and negative classes. With $C = 5$ categories, no single positive/negative split is canonical. For this reason, five one-versus-rest ROC curves are constructed: for each genre g , the predicted probability $\hat{\pi}_g(x)$ is used as a score, and the remaining four genres are pooled into the negative class. The resulting curves are displayed in Appendix B, Fig. 20. The per-class AUC values and their mean are reported as a summary of the overall multiclass discriminative performance.

VI. CONCLUSION

This work studied the problem of automatic music genre recognition from a high-dimensional set of acoustic descriptors. After a preprocessing stage that reduced redundancy and applied targeted logarithmic transformations, the data were analyzed both in an unsupervised and a supervised framework.

The unsupervised analysis revealed that the five musical genres do not form compact or well-separated clusters in the Euclidean spaces constructed from the retained PCA components. The hierarchical clustering with Ward’s method consistently preferred solutions with two or three groups, and the genre-wise silhouette values remained low throughout, indicating that the genre structure is not strongly encoded in the principal subspaces of the descriptor matrix.

In contrast, the supervised binary classifiers achieved high discriminative performance when distinguishing *Classical* from *Jazz*. All five models, including the full logistic model, the significance-based reductions, the stepwise AIC selection, and the ridge regularized classifier, produced test AUC values consistently above 0.94, confirming the strong linear separability of these two genres in the acoustic feature space. The stepwise AIC model (**ModAIC**) achieved the highest empirical test performance and was selected as the final binary classifier to generate the genre predictions for the unlabeled holdout file.

The multinomial extension to five genres introduced additional complexity, both statistically and computationally. The class probabilities were modeled through the softmax link function, and the Newton estimation procedure was shown to coincide with IRLS in this setting. The one-versus-rest evaluation framework confirmed that the discriminative performance varies across genres, with some categories being substantially easier to separate from the rest than others.

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APPENDIX A

USE OF GENERATIVE TOOLS

Generative tools were used only in narrowly delimited support tasks and not to replace the statistical analysis carried out in R. Their use was restricted to three situations. First, they were used to help implement formula-based code blocks whose practical implementation had not been explicitly detailed during the in-class coding sessions, while the underlying statistical definition itself remained standard and easily verifiable from course material or cited sources. This concerns in particular the heteroscedasticity screening score and the automatic choice of the number of clusters k through the silhouette criterion of Rousseeuw. These interventions are explicitly marked in `main.R` by dedicated comments.

Second, generative tools were used to suggest technical simplifications for pieces of code that were empirically found to be expensive in repeated execution. The main example is the replacement of some `ggplot` + `ggsave` rendering steps inside loops by a lighter `png()` + `base-plot()` strategy. This suggestion was not accepted blindly: it was kept only after checking that the output remained consistent with the intended graph and that the change effectively reduced execution time in practice.

Third, generative tools were used in the writing process of the report, mainly for translation and refinement. A first base structure was drafted in Spanish and Portuguese, and the tool was then used to help convert this material into a coherent IEEE-style LaTeX text in English. In this case, the tool was used for wording, organization, and local consistency improvements, not for producing numerical results.

The prompts were therefore formulated in a precise and technically constrained way. Representative examples include requests such as: “Implement in R a heteroscedasticity score for strictly positive variables using empirical quantile bins and Spearman correlation between group means and group standard deviations, and identify the sources that justify the construction”; “Write an R routine that, for Ward clustering on PCA scores, computes the mean silhouette width for $k = 2, \dots, 10$ and selects the maximizing value according to Rousseeuw’s definition”; “Suggest a lighter graphics-saving strategy for a loop where `ggplot` + `ggsave` is too slow, while preserving the intended output”; and “Rewrite this verified draft into concise IEEE LaTeX English without inventing results.”

All generated outputs were then validated manually before being retained. When the output involved a formula, the expression was checked against a cited source, against the equations written in the report, or against a direct derivation simple enough to verify locally.

APPENDIX B SUPPLEMENTARY DIAGNOSTICS

This appendix gathers additional descriptive material used in the preprocessing discussion.

A. Mean-to-Median Ratio

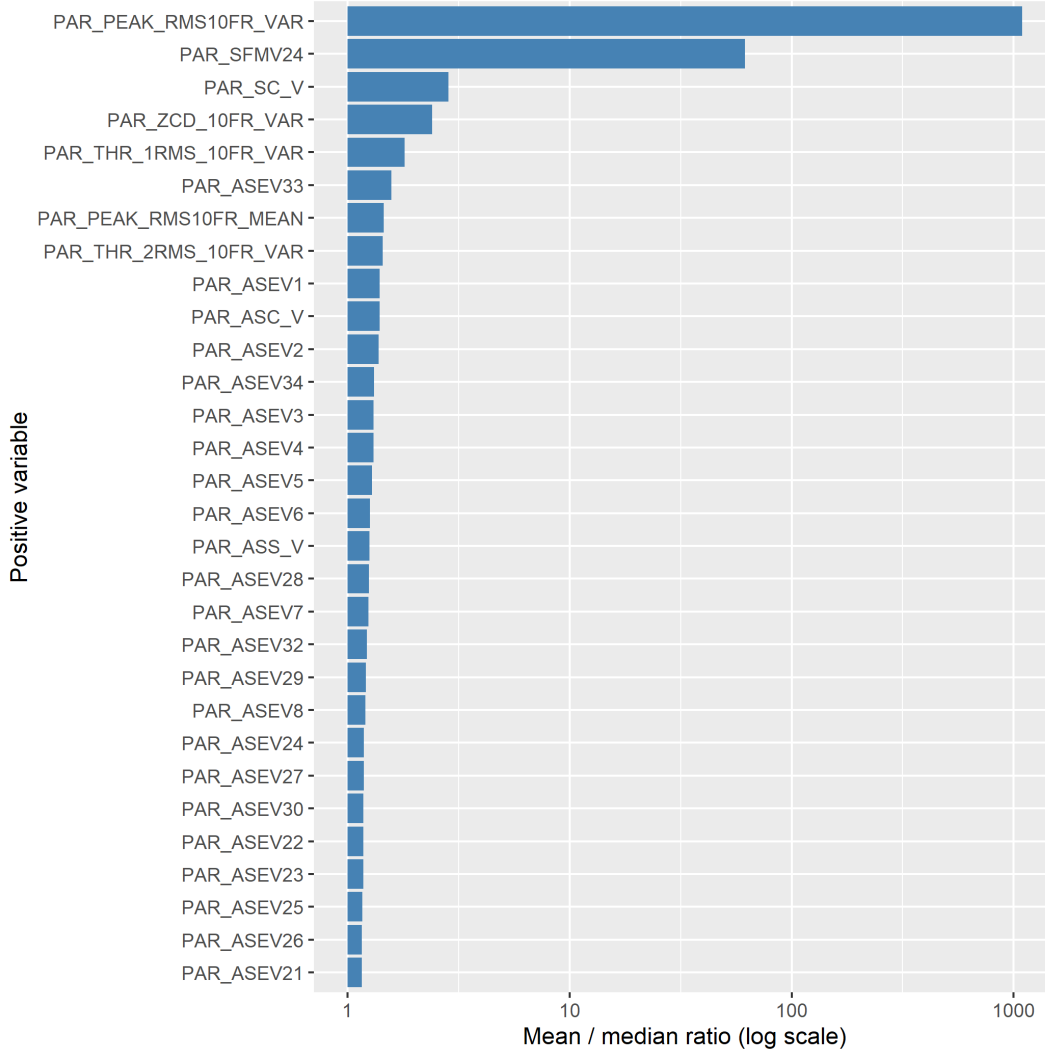


Fig. 4. Mean-to-median ratio for the positive variables considered as candidates for a logarithmic transformation. Larger values indicate stronger right-skewness.

B. Skewness

The skewness coefficient used in the report follows the standardized third central moment given by NIST/SEMATECH [3]:

$$g_1 = \frac{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})^3}{s^3}, \quad (9)$$

where $\bar{Y}$ is the sample mean and s is the sample standard deviation. Positive values indicate right-skewness, negative values indicate left-skewness, and values close to zero correspond to more symmetric distributions.

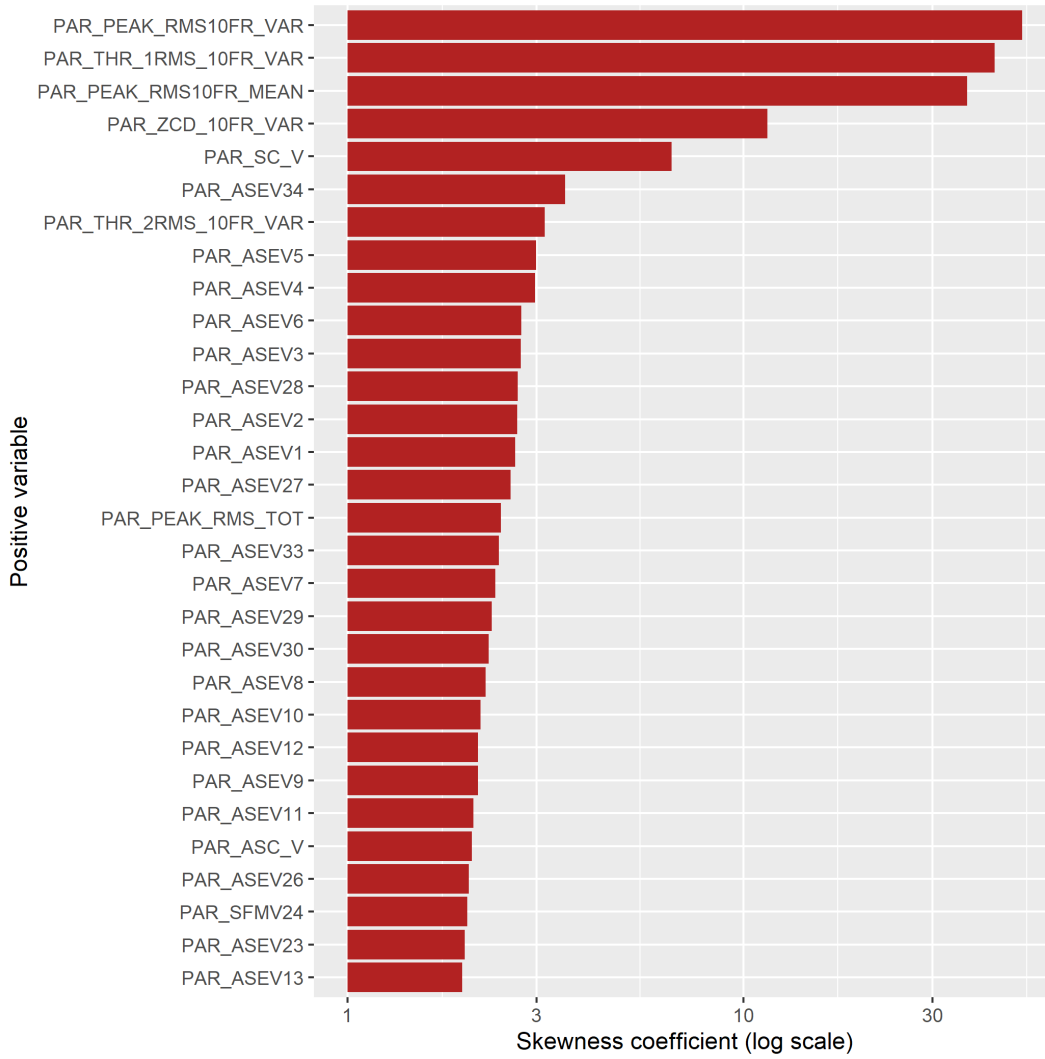


Fig. 5. Sample skewness for the variables considered as candidates for a logarithmic transformation. Larger positive values indicate stronger right-skewness.

C. Heteroscedasticity

Heteroscedasticity is important in the present analysis because, even when a variable is strongly right-skewed, a logarithmic transformation is especially relevant when the dispersion also tends to increase with the level of the measurements. Van den Berg et al. [1] note that logarithmic transformations are used in part to stabilize spread, while the NIST/SEMATECH *spread-location* diagnostic emphasizes the practical importance of examining how variability changes with location [5]. Following this idea, the implemented score is a descriptive screening tool: each strictly positive variable is partitioned into $K = 10$ empirical-quantile groups, the group means $\bar{X}_g$ and group standard deviations s_g are computed, and the monotone association between level and spread is summarized by Spearman's rank correlation [4],

$$H(X) = \rho_S((\bar{X}_g)_{g \in \mathcal{G}}, (s_g)_{g \in \mathcal{G}}), \quad (10)$$

where ρ_S is Spearman's coefficient and $\mathcal{G}$ denotes the set of groups for which both summaries are finite. Large positive values of $H(X)$ indicate a clear *spread-versus-level* pattern, meaning that higher average values are associated with higher within-group dispersion; for that reason, this score is used together with the mean-to-median ratio and skewness to identify plausible candidates for a logarithmic transformation. The corresponding ranking of variables is shown in Fig. 6.

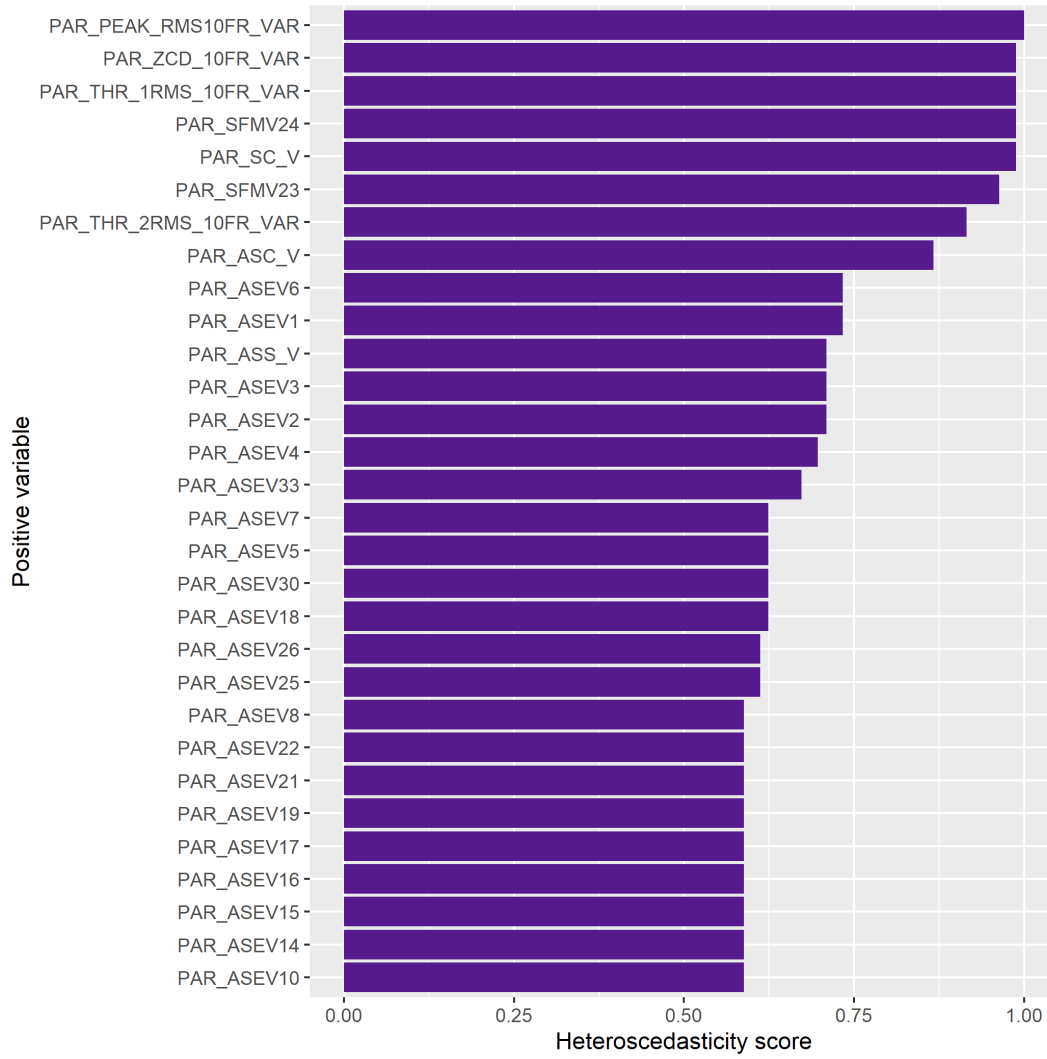


Fig. 6. Heteroscedasticity score for the positive variables considered as candidates for a logarithmic transformation. Larger positive values indicate a stronger spread-versus-level pattern.

D. Log-Normality

For the variables retained simultaneously by the asymmetry and heteroscedasticity screens, we finally examine whether the logarithmic transformation produces a distribution closer to normality. The set of common candidate variables is listed in Table V. We then apply the Shapiro–Wilk test to $\log(X)$ and rank these variables by the resulting W statistic, where values closer to 1 indicate a better agreement with normality. In basic terms, the **variable** column gives the name of the candidate predictor, **shapiro_w** is the Shapiro–Wilk statistic computed on $\log(X)$, and **p_value** is the corresponding significance level for the null hypothesis that $\log(X)$ is normally distributed. The values reported in Table VI are obtained by applying the test to each positive variable that was retained by the previous screening criteria.

TABLE V
VARIABLES RETAINED SIMULTANEOUSLY BY THE ASYMMETRY AND HETEROSCEDASTICITY SCREENING CRITERIA.

#	Variable
1	PAR_PEAK_RMS10FR_VAR
2	PAR_SFMV24
3	PAR_SC_V
4	PAR_ZCD_10FR_VAR
5	PAR_THR_1RMS_10FR_VAR
6	PAR_ASEV33
7	PAR_THR_2RMS_10FR_VAR
8	PAR_ASEV1
9	PAR_ASC_V
10	PAR_ASEV2
11	PAR_ASEV3
12	PAR_ASEV4
13	PAR_ASEV5
14	PAR_ASEV6
15	PAR_ASEV7
16	PAR_ASEV8
17	PAR_ASEV30
18	PAR_ASEV26

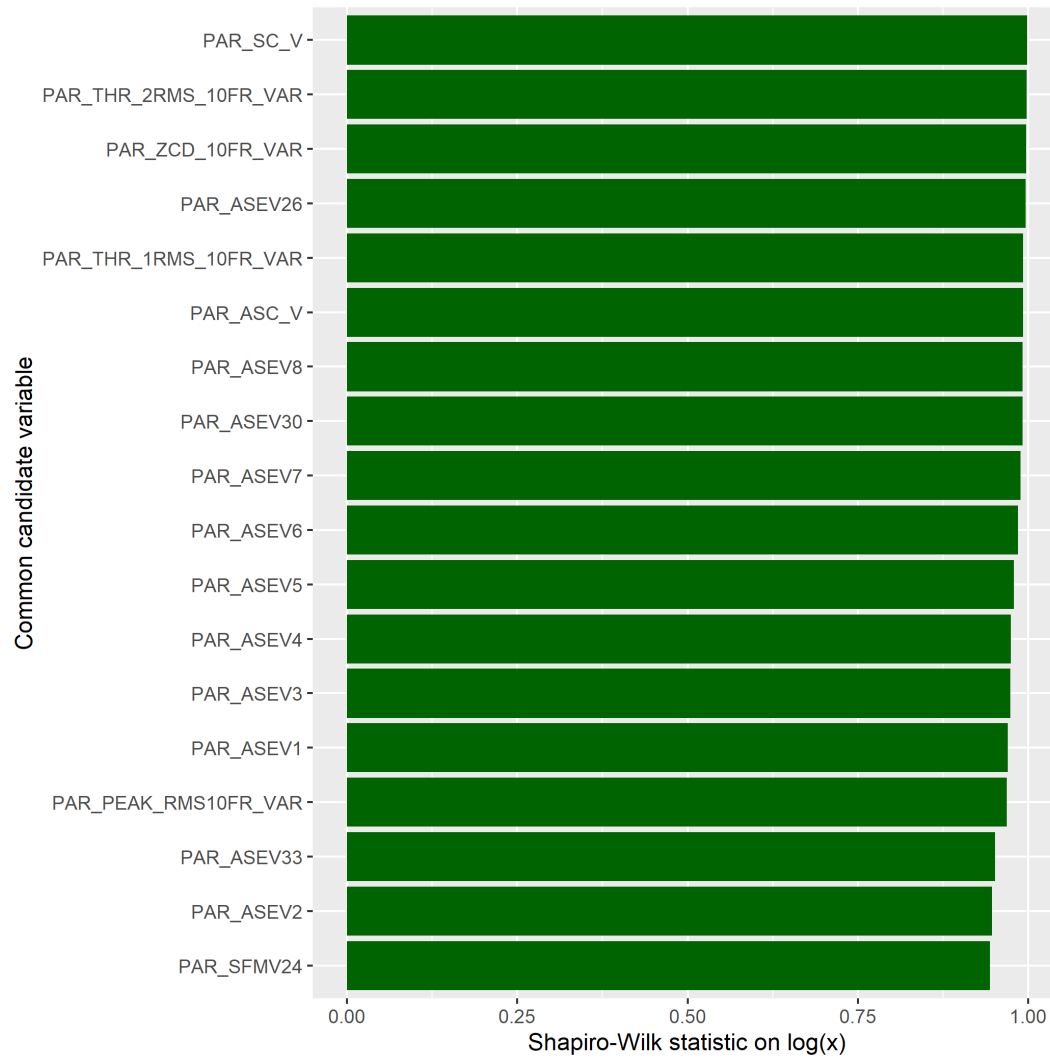


Fig. 7. Shapiro-Wilk W statistic computed on $\log(X)$ for the common candidate variables. Larger values indicate that the log-transformed variable is closer to a normal distribution.

TABLE VI
SHAPIRO-WILK RESULTS ON $\log(X)$ FOR THE COMMON CANDIDATE VARIABLES.

Variable	W	p -value
PAR_SC_V	0.9984029	6.06×10^{-5}
PAR_THR_2RMS_10FR_VAR	0.9977017	7.98×10^{-7}
PAR_ZCD_10FR_VAR	0.9969759	1.70×10^{-8}
PAR_ASEV26	0.9961615	4.01×10^{-10}
PAR_THR_1RMS_10FR_VAR	0.9922652	7.13×10^{-16}
PAR_ASC_V	0.9922296	6.47×10^{-16}
PAR_ASEV8	0.9916860	1.51×10^{-16}
PAR_ASEV30	0.9913730	6.72×10^{-17}
PAR_ASEV7	0.9888169	1.77×10^{-19}
PAR_ASEV6	0.9853344	2.19×10^{-22}
PAR_ASEV5	0.9790100	1.45×10^{-26}
PAR_ASEV4	0.9746038	6.10×10^{-29}
PAR_ASEV3	0.9737920	2.41×10^{-29}
PAR_ASEV1	0.9698754	3.67×10^{-31}
PAR_PEAK_RMS10FR_VAR	0.9688232	1.28×10^{-31}
PAR_ASEV33	0.9510785	6.11×10^{-38}
PAR_ASEV2	0.9469831	3.93×10^{-39}
PAR_SFMV24	0.9436739	4.84×10^{-40}

E. PCA Variable Space

The following tables summarize the main diagnostics used to interpret the variable space of the PCA after preprocessing and standardization of the 165 retained predictors.

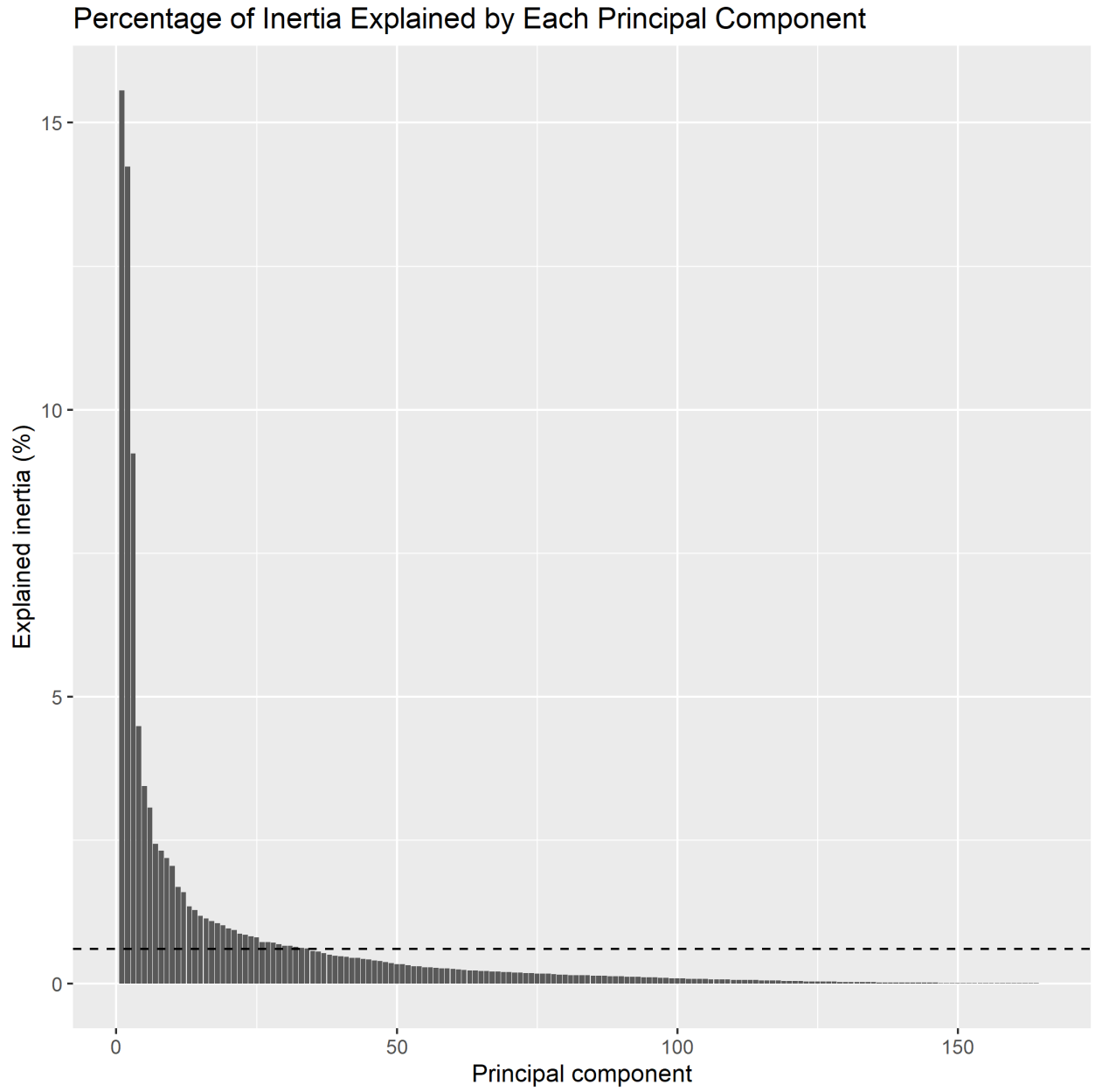


Fig. 8. Percentage of inertia explained by each principal component. The dashed horizontal line marks the average contribution 100/165.

TABLE VII
CUMULATIVE INERTIA AND MEAN INDIVIDUAL QUALITY OF REPRESENTATION FOR SELECTED PCA SUBSPACES.

Dimensions	Cumulative inertia (%)	Mean individual $\cos^2$
2	29.7942	0.2888
4	43.5192	0.4283
6	50.0195	0.4970
10	59.0076	0.5900
34	81.6010	0.8064

TABLE VIII
TEN VARIABLES WITH THE LARGEST CONTRIBUTIONS TO PC1.

Variable	Contrib.	Corr.	$\cos^2$ axis	$\cos^2(1, 2)$	$\cos^2(1:4)$
PAR_ASE14	2.512	-0.803	0.645	0.647	0.661
PAR_ASE15	2.335	-0.774	0.600	0.600	0.604
PAR_ASE13	2.266	-0.763	0.582	0.597	0.627
PAR_ASS	2.180	0.748	0.560	0.564	0.759
PAR_MFCC1	2.134	-0.740	0.548	0.581	0.898
PAR_ASE17	2.083	-0.731	0.535	0.552	0.577
PAR_SFM11	2.023	0.721	0.520	0.703	0.705
PAR_ASE18	2.020	-0.720	0.519	0.532	0.562
PAR_ASE12	2.001	-0.717	0.514	0.553	0.618
PAR_ASC_V	1.986	0.714	0.510	0.701	0.729

TABLE IX
TEN VARIABLES WITH THE LARGEST CONTRIBUTIONS TO PC2.

Variable	Contrib.	Corr.	$\cos^2$ axis	$\cos^2(1, 2)$	$\cos^2(1:4)$
PAR_3RMS_TCD	1.659	0.624	0.390	0.424	0.469
PAR_ASE7	1.648	-0.622	0.387	0.414	0.686
PAR_ASE8	1.641	-0.621	0.385	0.385	0.580
PAR_ASEV22	1.617	0.616	0.380	0.392	0.493
PAR_3RMS_TCD_10FR_MEAN	1.569	0.607	0.369	0.405	0.448
PAR_ASEV26	1.564	0.606	0.367	0.372	0.493
PAR_ASEV27	1.515	0.596	0.356	0.373	0.450
PAR_ASEV24	1.500	0.594	0.352	0.354	0.475
PAR_ASEV25	1.488	0.591	0.349	0.351	0.480
PAR_ASEV23	1.468	0.587	0.345	0.349	0.470

TABLE X
TEN VARIABLES WITH THE LARGEST CONTRIBUTIONS TO PC3.

Variable	Contrib.	Corr.	$\cos^2$ axis	$\cos^2(1, 2)$	$\cos^2(1:4)$
PAR_ASE23	3.612	-0.742	0.551	0.085	0.670
PAR_ASE25	3.570	-0.738	0.544	0.165	0.747
PAR_ASE24	3.364	-0.716	0.513	0.122	0.677
PAR_ASE22	3.223	-0.701	0.491	0.138	0.650
PAR_1RMS_TCD	3.067	-0.684	0.467	0.294	0.762
PAR_ASE26	2.966	-0.672	0.452	0.240	0.711
PAR_ASE27	2.582	-0.627	0.393	0.284	0.681
PAR_2RMS_TCD	2.516	-0.619	0.383	0.384	0.768
PAR_ASC	2.466	-0.613	0.376	0.373	0.778
PAR_2RMS_TCD_10FR_MEAN	2.431	-0.609	0.370	0.364	0.735

TABLE XI
TEN VARIABLES WITH THE LARGEST CONTRIBUTIONS TO PC4.

Variable	Contrib.	Corr.	$\cos^2$ axis	$\cos^2(1, 2)$	$\cos^2(1:4)$
PAR_ASE34	3.761	-0.528	0.278	0.017	0.573
PAR_SFMV22	3.472	0.507	0.257	0.156	0.426
PAR_ASE5	3.367	0.499	0.249	0.411	0.757
PAR_ASE6	3.363	0.499	0.249	0.419	0.746
PAR_ASE4	2.812	0.456	0.208	0.402	0.722
PAR_ASE33	2.808	-0.456	0.208	0.083	0.459
PAR_SFM22	2.688	-0.446	0.199	0.088	0.425
PAR_ASE7	2.662	0.444	0.197	0.414	0.686
PAR_SFMV21	2.654	0.443	0.197	0.305	0.501
PAR_ASE3	2.113	0.396	0.157	0.387	0.665

TABLE XII
VARIABLES BEST REPRESENTED ON THE FIRST PRINCIPAL PLANE.

Variable	$\cos^2(1,2)$
PAR_SC_V	0.780
PAR_SFM15	0.719
PAR_SFM11	0.703
PAR_SFM12	0.703
PAR_ASC_V	0.701
PAR_SFM14	0.691
PAR_SFM16	0.676
PAR_SFM10	0.674
PAR_SFM13	0.663
PAR_ASE14	0.647
PAR_SFM9	0.635
PAR_ASE15	0.600
PAR_ASE13	0.597
PAR_SFM17	0.596
PAR_MFCC1	0.581

TABLE XIII
VARIABLES BEST REPRESENTED ON THE FIRST FOUR PRINCIPAL COMPONENTS.

Variable	$\cos^2(1:4)$
PAR_MFCC1	0.898
PAR_ZCD	0.841
PAR_SC_V	0.815
PAR_SC	0.799
PAR_ASC	0.778
PAR_2RMS_TCD	0.768
PAR_1RMS_TCD	0.762
PAR_ASS	0.759
PAR_ASE5	0.757
PAR_SFM15	0.752
PAR_ASE25	0.747
PAR_ASE6	0.746
PAR_SFM16	0.746
PAR_2RMS_TCD_10FR_MEAN	0.735
PAR_ASC_V	0.729

TABLE XIV
COUNTS OF WELL-REPRESENTED VARIABLES IN THE PCA VARIABLE SPACE.

Criterion	Count
Variables with $\cos^2(1,2) \geq 0.5$	24
Variables with $\cos^2(1,2) \geq 0.7$	5
Variables with $\cos^2(1:4) \geq 0.5$	69
Variables with $\cos^2(1:4) \geq 0.7$	20

F. Silhouette Criterion

For the clustering step, the number of clusters is selected by maximizing the average silhouette width. For an observation x_i , the silhouette index compares the average dissimilarity to its own cluster with the smallest average dissimilarity to another cluster [11]:

$$\text{sil}(x_i) = \frac{b_i - a_i}{\max(a_i, b_i)}. \quad (11)$$

Here, a_i is the average distance from x_i to the observations of its own cluster, whereas b_i is the smallest average distance from x_i to another cluster. Larger values indicate that the observation is better assigned to its cluster and more clearly separated from the others.

G. Silhouette Plots

The following figures display the silhouette plots obtained for Ward clustering and for the true genre partition, for each retained PCA dimension and for both preprocessing modes.

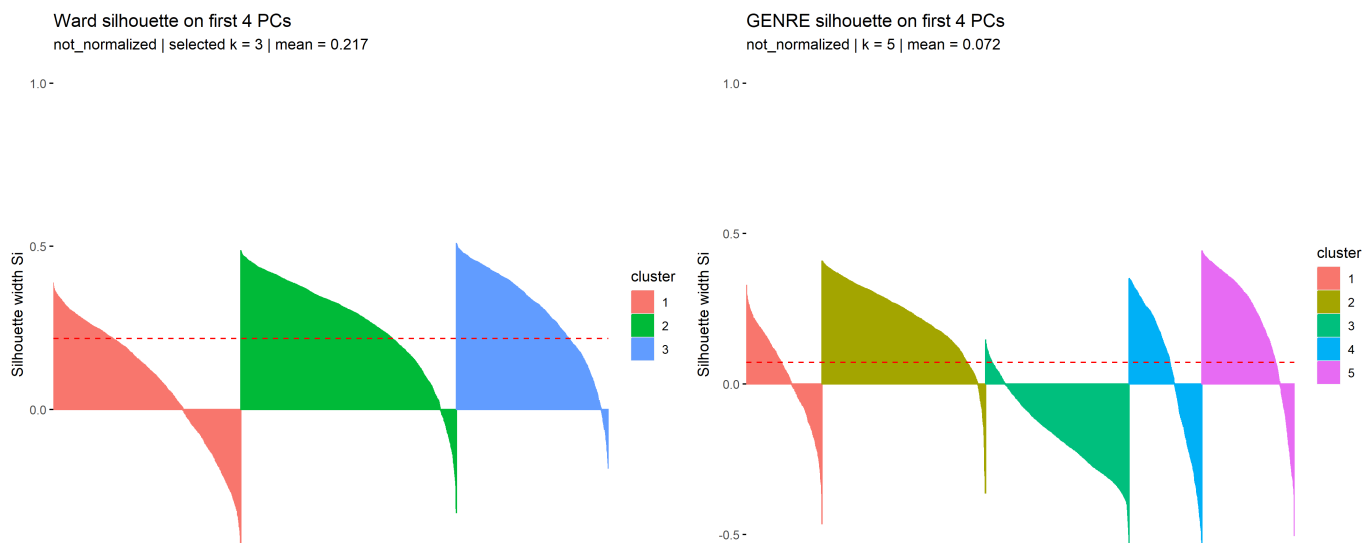


Fig. 9. Silhouette plots for the non-normalized PCA scores with $d = 4$ retained components.

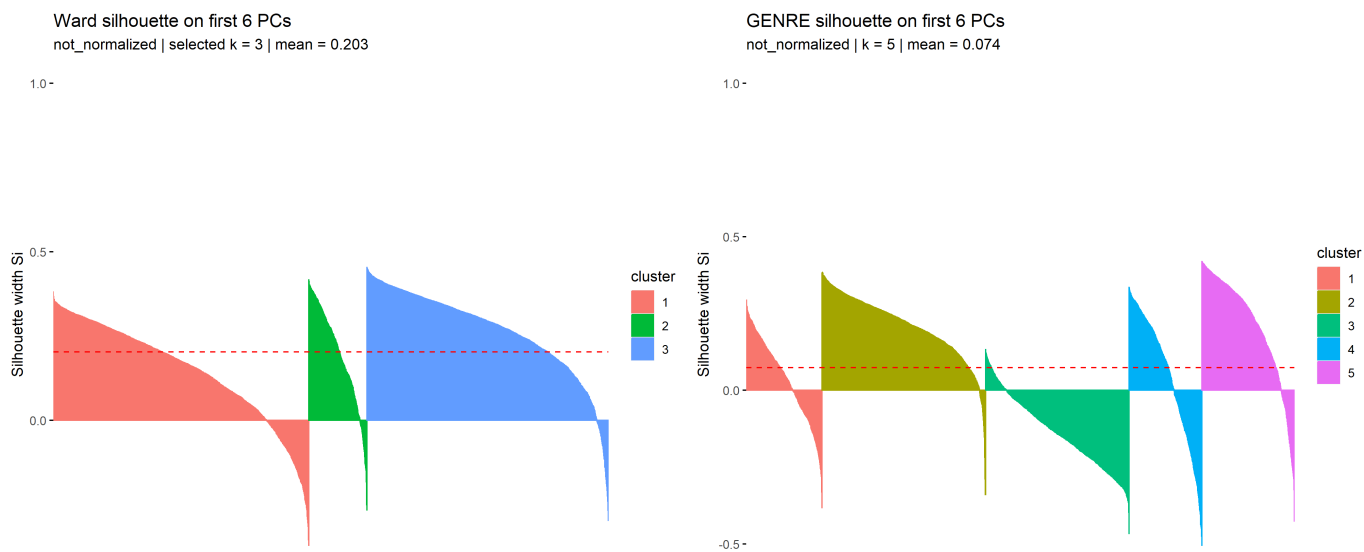


Fig. 10. Silhouette plots for the non-normalized PCA scores with $d = 6$ retained components.

Ward silhouette on first 10 PCs
not_normalized | selected k = 2 | mean = 0.164

GENRE silhouette on first 10 PCs
not_normalized | k = 5 | mean = 0.077

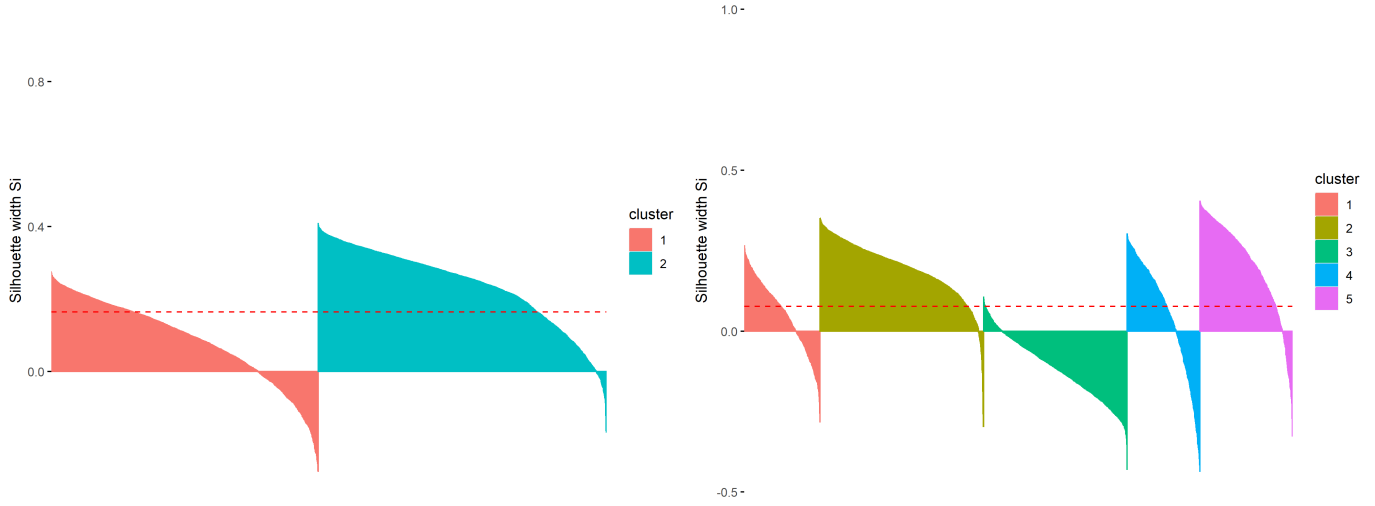


Fig. 11. Silhouette plots for the non-normalized PCA scores with $d = 10$ retained components.

Ward silhouette on first 34 PCs
not_normalized | selected k = 2 | mean = 0.121

GENRE silhouette on first 34 PCs
not_normalized | k = 5 | mean = 0.056

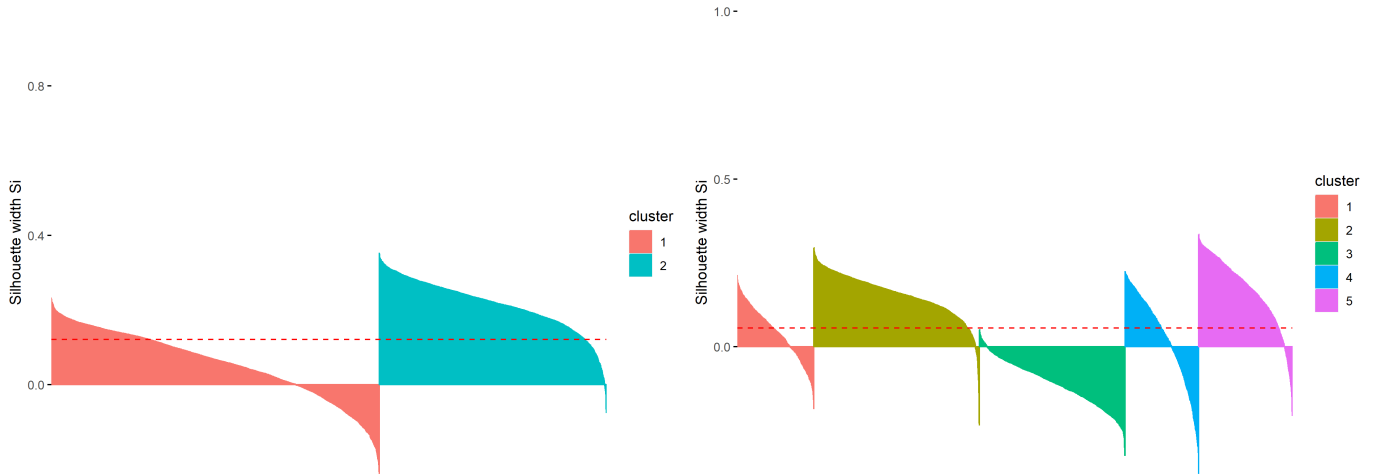


Fig. 12. Silhouette plots for the non-normalized PCA scores with $d = 34$ retained components.

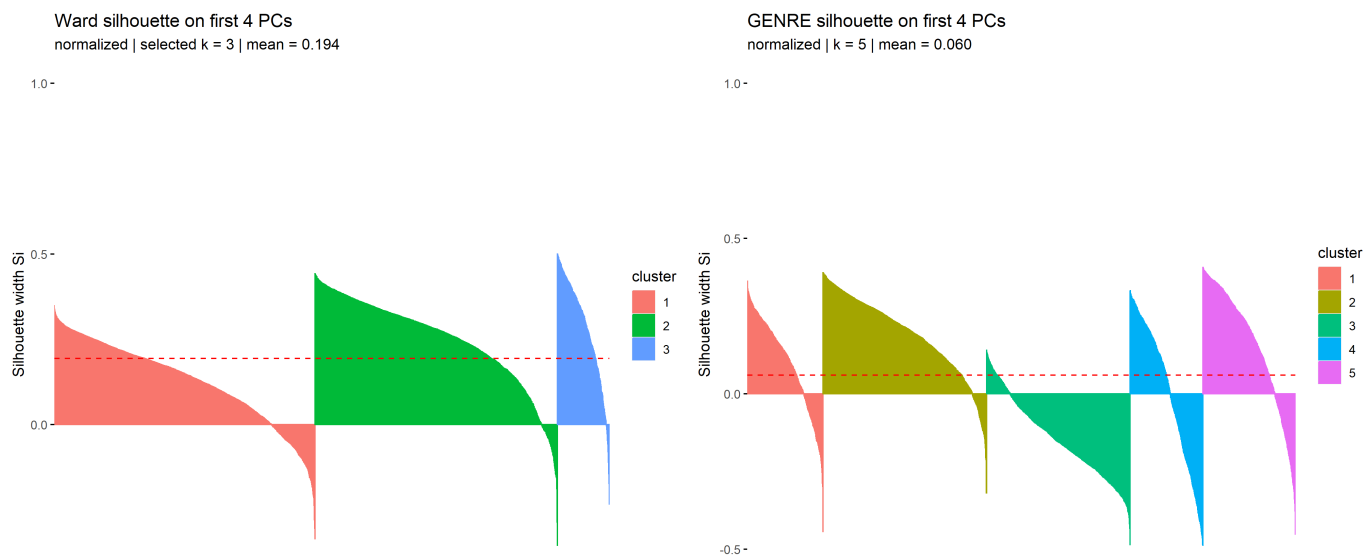


Fig. 13. Silhouette plots for the normalized PCA scores with $d = 4$ retained components.

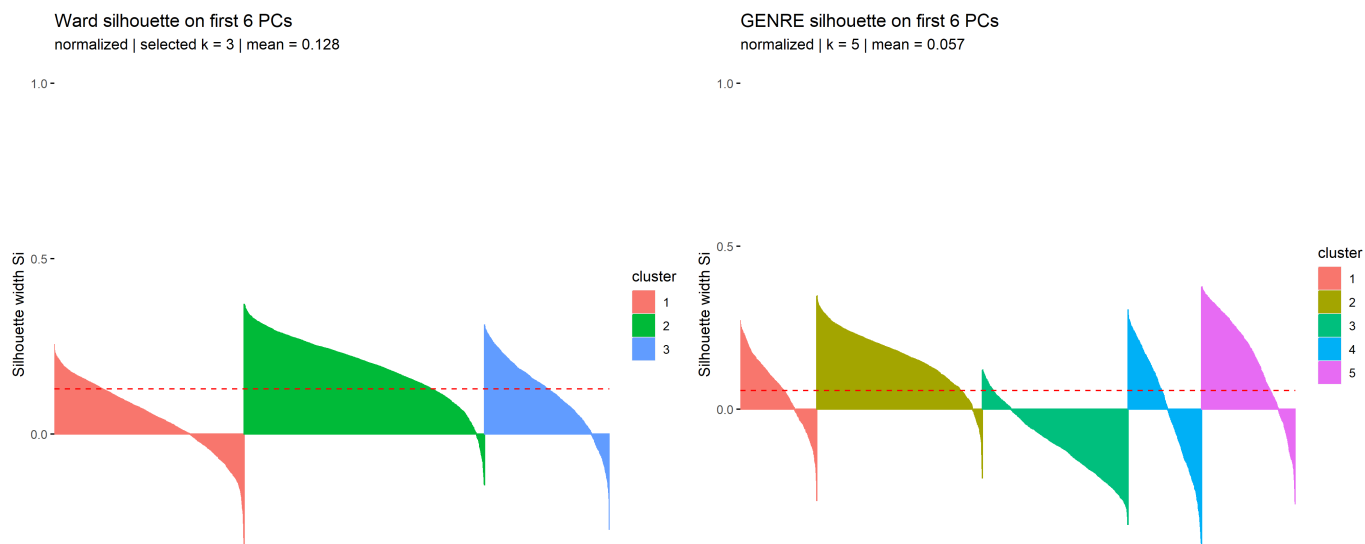


Fig. 14. Silhouette plots for the normalized PCA scores with $d = 6$ retained components.

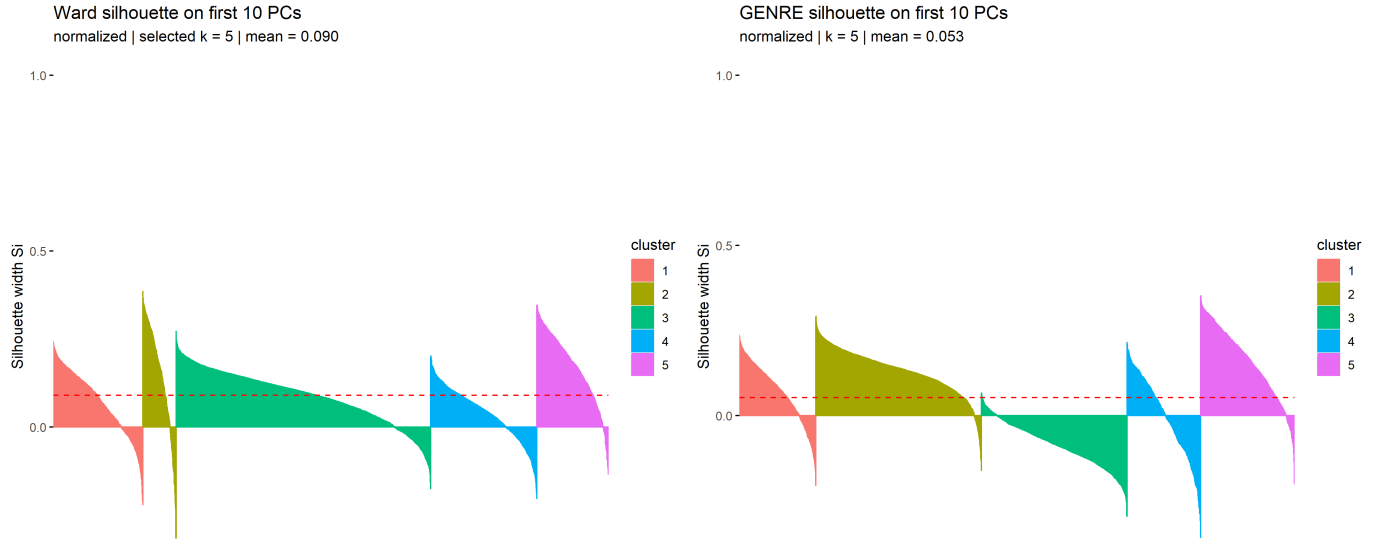


Fig. 15. Silhouette plots for the normalized PCA scores with $d = 10$ retained components.

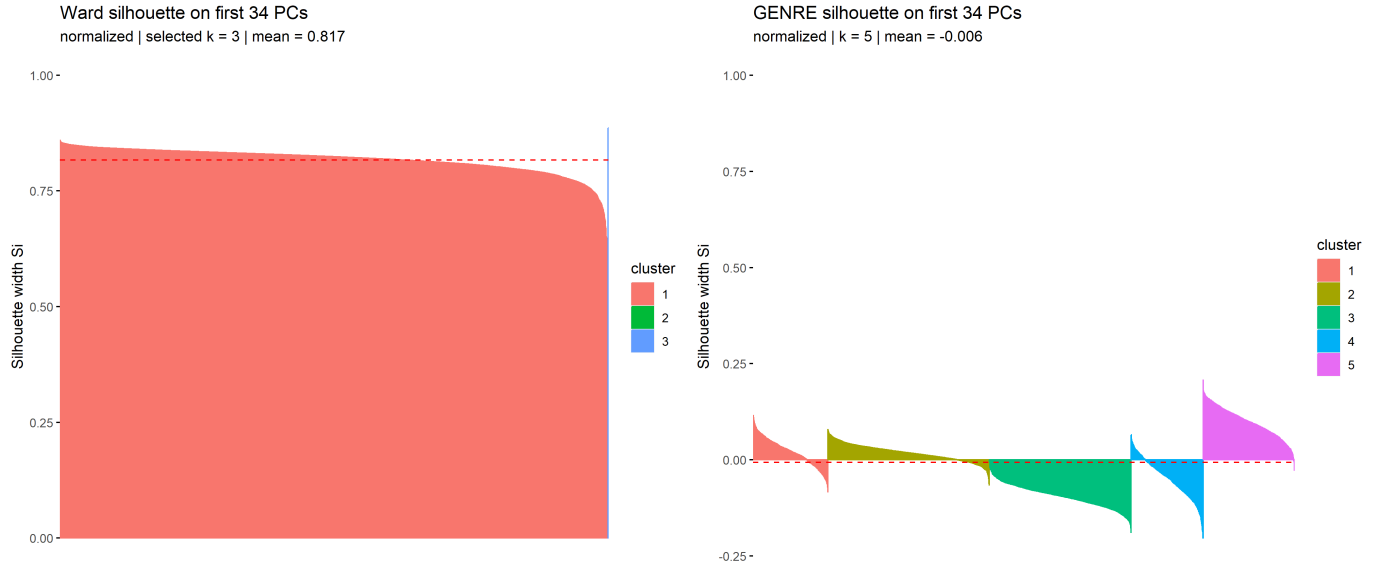


Fig. 16. Silhouette plots for the normalized PCA scores with $d = 34$ retained components.

H. Coefficient Significance in Logistic Regression

In the binary logistic models of Part II, the significance of each predictor is assessed through the Wald test for an individual regression coefficient [6], [7]. If $\pi(x) = \mathbb{P}(Y = 1 \mid X = x)$, the fitted model is

$$\log\left(\frac{\pi(x)}{1 - \pi(x)}\right) = \beta_0 + \sum_{j=1}^p \beta_j x_j. \quad (12)$$

For each predictor x_j , the implemented null hypothesis is

$$H_0 : \beta_j = 0. \quad (13)$$

Under the large-sample normal approximation for the maximum likelihood estimator,

$$Z_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \approx \mathcal{N}(0, 1) \quad \text{under } H_0, \quad (14)$$

or equivalently

$$W_j = Z_j^2 = \frac{\hat{\beta}_j^2}{\text{SE}(\hat{\beta}_j)^2} \approx \chi_1^2. \quad (15)$$

Hence the column $\text{Pr}(> |z|)$ returned by `summary(glm(...))` is the two-sided p-value

$$p_j = 2 \mathbb{P}(Z \geq |Z_j|) = 2 [1 - \Phi(|Z_j|)], \quad Z \sim \mathcal{N}(0, 1). \quad (16)$$

In the present implementation, the reduced model **Mod1** retains the variables such that $p_j < 0.05$, whereas **Mod2** retains the variables such that $p_j < 0.20$.

I. Stepwise AIC Selection

The model **ModAIC** of Part II is obtained from the full logistic model **ModT** by applying `stepAIC(ModT)` from the **MASS** package [8], [9]. The criterion minimized along the search is Akaike's Information Criterion,

$$\text{AIC} = -2 \ell(\hat{\beta}) + k q, \quad (17)$$

where $\ell(\hat{\beta})$ is the maximized log-likelihood of the candidate model, q is the number of estimated parameters, and $k = 2$ for the classical AIC. Since the code uses `stepAIC(ModT)` with default arguments, the search starts from the full logistic model and removes terms step by step whenever this decreases the AIC. At each step, among the admissible one-term deletions, the selected move is the one that gives the smallest AIC, and the procedure stops when no further deletion improves the criterion. Formally, if $\mathcal{M}_0$ is the current model and $\mathcal{N}(\mathcal{M}_0)$ is the set of candidate one-step submodels explored by the algorithm, the update rule can be summarized as

$$\mathcal{M}_{t+1} = \arg \min_{\mathcal{M} \in \mathcal{N}(\mathcal{M}_t)} \text{AIC}(\mathcal{M}), \quad (18)$$

provided that this minimum is strictly smaller than $\text{AIC}(\mathcal{M}_t)$. The resulting model therefore balances fit and parsimony through the AIC penalty rather than through coefficient-wise significance thresholds.

J. Ridge Regularization

Following the regularization framework presented in [10], the motivation for ridge regression can first be introduced from the classical linear regression setting. In ordinary least squares, the objective is to minimize the residual sum of squares:

$$\text{SCR}(\theta) = \sum_{i=1}^n (y_i - x_i^\top \theta)^2. \quad (19)$$

As shown in Equation (19), the estimator seeks the coefficient vector θ that best fits the training data. However, when the number of predictors increases, the model becomes more flexible. This can reduce bias, because the model has greater capacity to adapt to the observed structure, but it can also increase variance, since small perturbations in the training sample may lead to substantially different coefficient estimates.

Ridge regression addresses this issue by replacing the ordinary least-squares criterion with a penalized criterion [10]:

$$\text{SCR}(\theta) + \kappa \sum_{j=1}^p \theta_j^2. \quad (20)$$

Equivalently, the ridge estimator is defined as

$$\hat{\theta}_{\text{ridge}} = \arg \min_{\theta} \left[\sum_{i=1}^n (y_i - x_i^\top \theta)^2 + \kappa \sum_{j=1}^p \theta_j^2 \right]. \quad (21)$$

In Equations (20) and (21), the term

$$\kappa \sum_{j=1}^p \theta_j^2 \quad (22)$$

penalizes excessively large coefficients. The parameter $\kappa \geq 0$ controls the strength of the regularization. When $\kappa = 0$, the non-regularized estimator is recovered. As κ increases, the penalty becomes stronger and the estimated coefficients are progressively shrunk toward zero.

From the perspective of the bias–variance trade-off, ridge regularization introduces additional bias in order to reduce the variance of the estimator. This is particularly relevant in the presence of correlated predictors, because ridge tends to distribute the effect more smoothly among related variables instead of producing unstable coefficient estimates.

Therefore, ridge is not primarily used for variable selection, but rather to stabilize the estimation process, control the effective complexity of the model, and improve generalization performance.

In the logistic regression context considered in this project, the same regularization principle is applied by penalizing the negative log-likelihood. If $\ell(\beta)$ denotes the log-likelihood of the logistic model, the ridge logistic estimator is obtained by solving

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \left[-\ell(\beta) + \kappa \sum_{j=1}^p \beta_j^2 \right]. \quad (23)$$

Thus, Equation (23) extends the ridge principle described in [10] to the binary classification problem. In this setting, the regularization term reduces the magnitude of the logistic regression coefficients, improves numerical stability, and limits overfitting when the predictor space is high-dimensional.

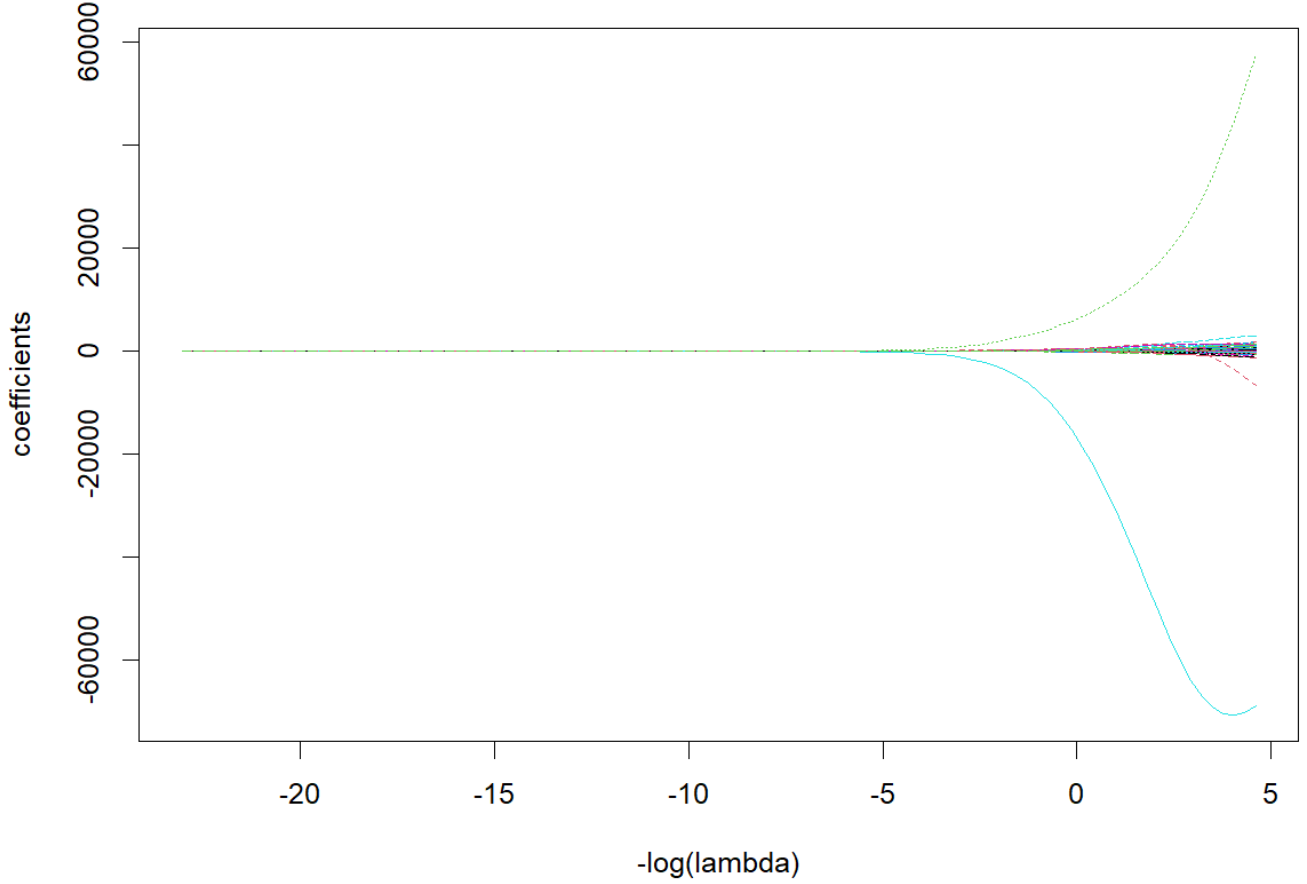


Fig. 17. Ridge coefficient paths as functions of $-\log(\lambda)$. Moving to the right corresponds to weaker penalization and larger coefficient magnitudes.

K. ROC Curves

The following figures report the ROC comparisons produced in Part II, first for the maximum-likelihood logistic models and then after adding the ridge classifier.

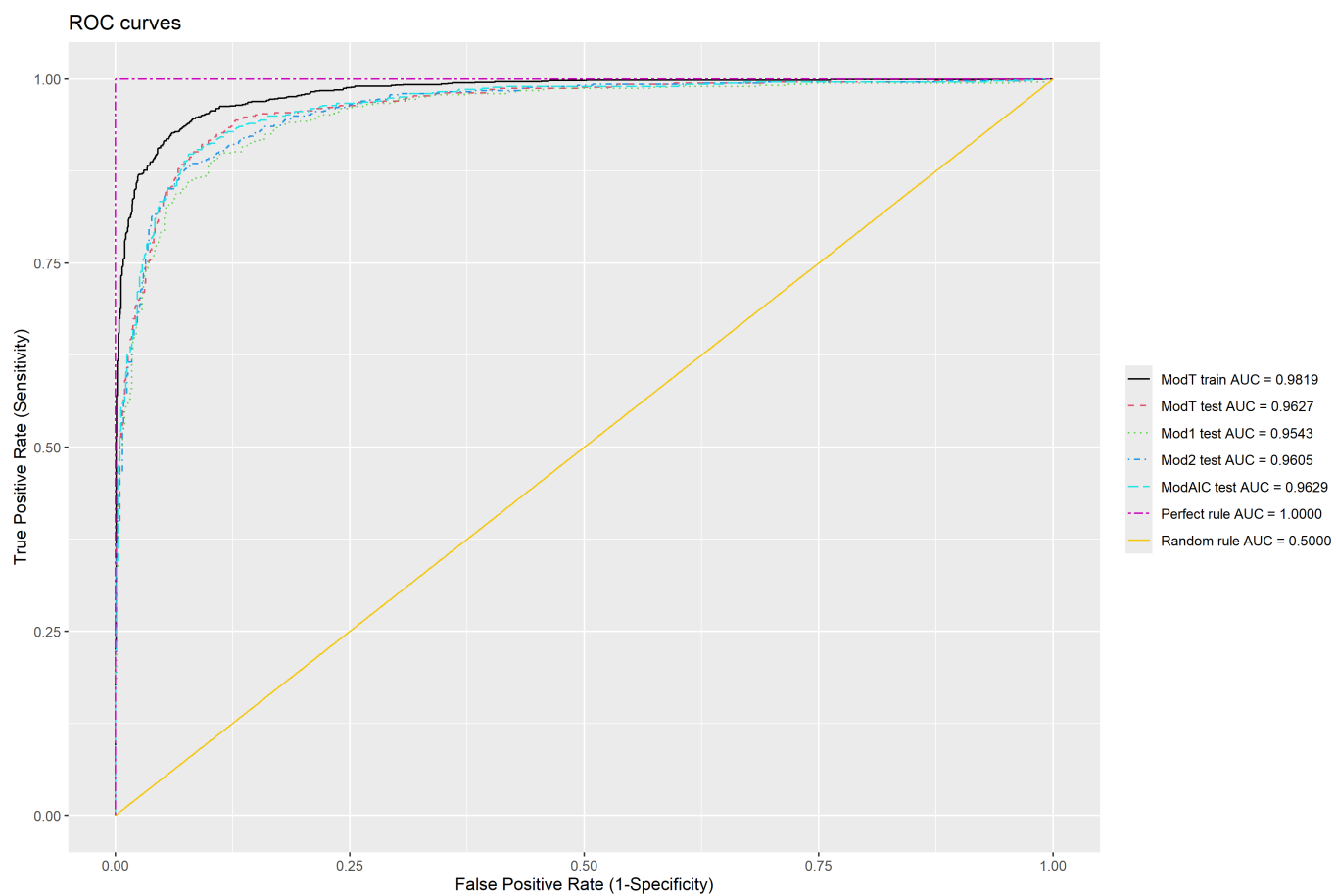


Fig. 18. ROC curves for the full logistic model, the significance-based reduced models, and the stepwise AIC model.

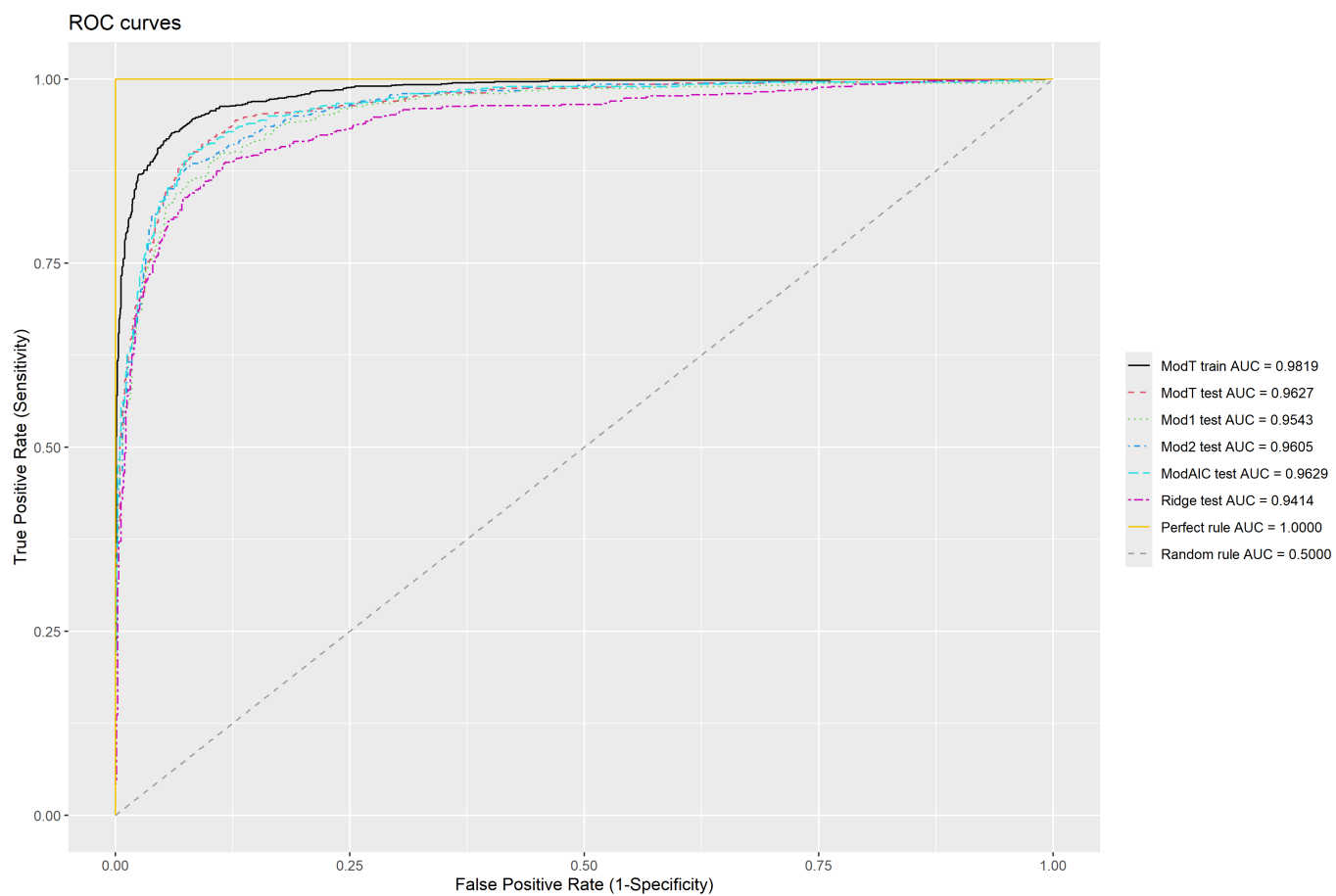


Fig. 19. ROC curves after adding the ridge logistic classifier to the previous comparison.

L. One-versus-Rest ROC Curves (Multinomial)

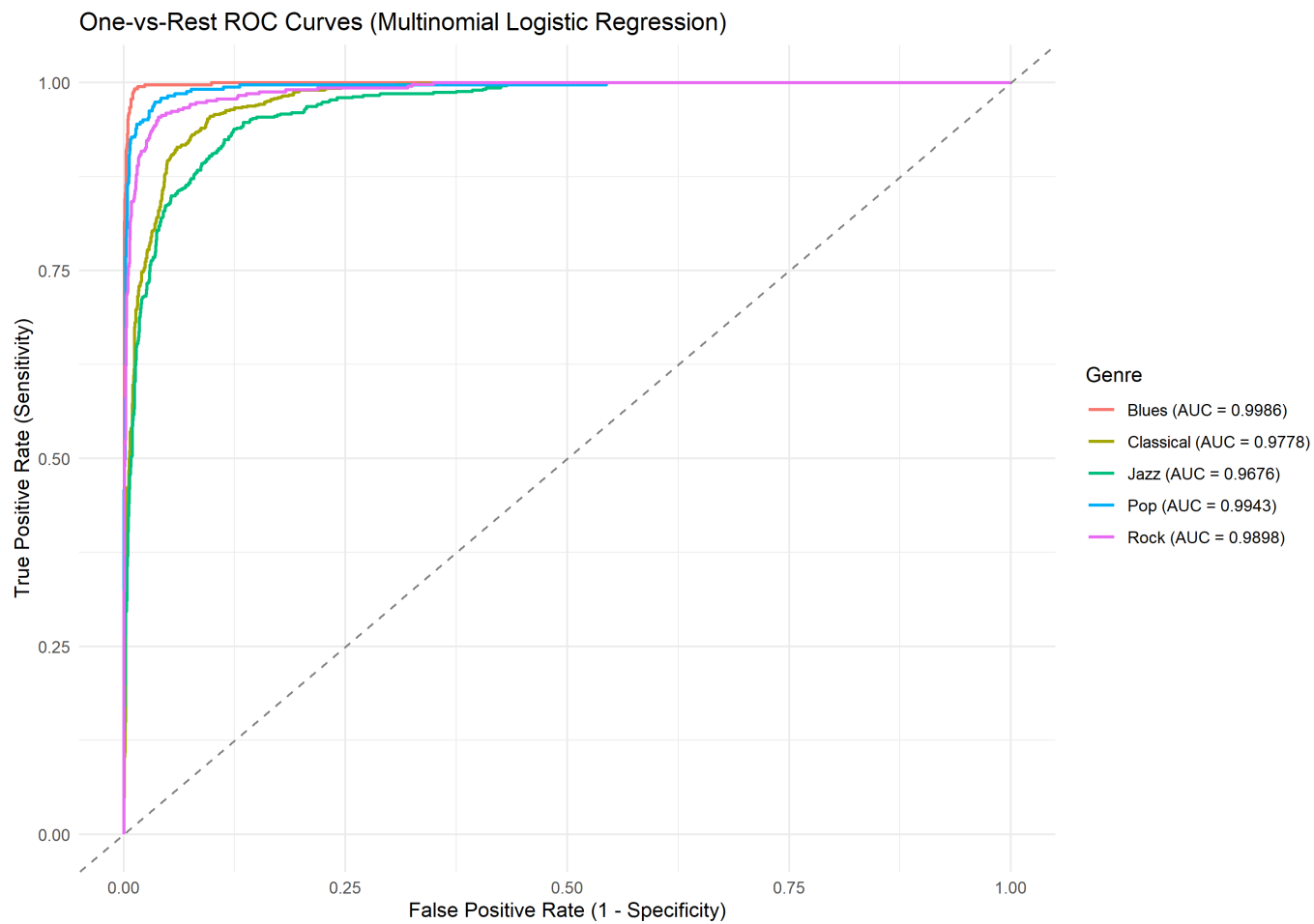


Fig. 20. One-versus-rest ROC curves for the five-class multinomial logistic model. Each curve evaluates the discrimination of one genre against the pooled remaining four genres.